

BEYOND THE BASE: EXPONENT ALLOCATION IN ROPE

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ABSTRACT

A finite rotary position embedding (RoPE) table chooses both a frequency range and an allocation of rotary pairs within that range. We study interior exponent allocation as a design freedom that organizes the positional basis, shapes what a model learns, and improves length generalization. Paired training at identical sampled endpoints isolates its effect: changing only 30 interior exponents improves extrapolation loss at $2\times$, $4\times$, and $8\times$ the training length in all three seeds. Full sine-cosine subspace analysis reveals how finite-window overlap concentrates positional directions, while crossed evaluations show that weights co-adapt to their allocation. Retargeting the runtime range reverses the ordering, connecting allocation to the scales at which a learned basis is used. We turn this perspective into analytic constructions and evaluate them across training, continuation, adaptation, and frozen deployment. EVQ-Cosh reduces 16K perplexity from 138.8 to 95.6 in three-seed MLA training; intermediate-band redistribution improves five-task natural-QA F1 from 21.62 to 25.44 on a frozen OLMo checkpoint. Together, the controlled interventions, geometric analysis, and model results establish interior allocation as a consequential and usable dimension of RoPE design.

1 INTRODUCTION

A RoPE frequency table determines the positional scales available to an attention head. Its standard geometric construction is usually specified by a base, which sets the sampled range of those scales (Su et al., 2024; Liu et al., 2024; Xu et al., 2024). Within any such range, a finite number of rotary pairs must still be placed. Their placement determines which positional functions the model receives and how densely different parts of the range are represented. This interior allocation is an additional design choice, even when the fastest and slowest frequencies are held exactly fixed.

Frequency design already has substantial practical value. Learned schedules improve language modeling, and nonuniform frequency transformations support context extension (Karypis et al., 2026; Wang et al., 2026; Peng et al., 2024; Ding et al., 2024; Tian et al., 2026). These developments motivate a more precise question: *what does the allocation of interior frequencies contribute, and how does that contribution pass through the positional basis, learned weights, and model behavior?* Answering it requires separating the range a table covers from the placement of its samples and their assignment to learned query-key coordinates.

We combine controlled interventions, full-pair geometry, and concrete designs. Figure 1 starts from paired 151.9M training: changing only 30 interior exponents improves extended-length loss in every seed. Runtime range retargeting reverses the ordering, and crossed evaluations reveal strong weight-table compatibility. The geometric analysis characterizes the positional directions being allocated; training and frozen-model experiments show how that freedom becomes useful.

Each rotary pair supplies a complete sine-cosine space. Canonical overlap and an exact effective-rank identity show how distinct frequencies can concentrate into shared positional directions over a finite window. Learned projections associate these directions with coordinate slots. This connects the supplied basis to its use by a model and motivates two practical settings: constructing a table that the model can learn, and adjusting the native table of an existing checkpoint. Analytic allocations, complete-answer generation, low-dimensional profile transport, and natural QA provide complementary evidence of this design freedom’s value.

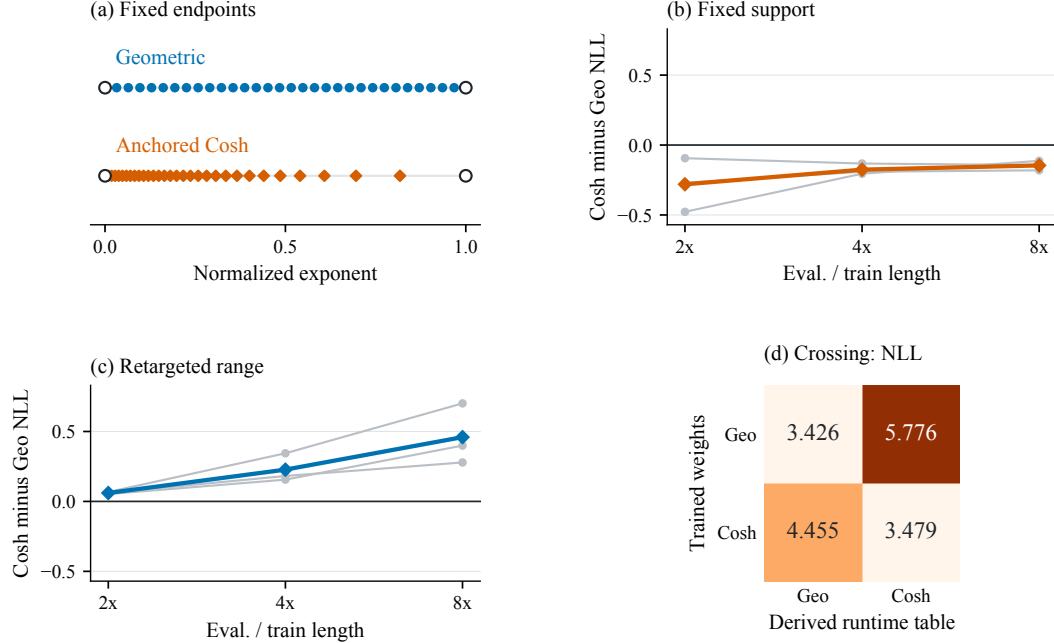


Figure 1: **Interior allocation shapes learning and its use at longer ranges.** (a) Two allocations share both sampled endpoints and 32 rotary pairs. (b,c) Cosh-minus-Geo tail NLL at $2 \times / 4 \times / 8 \times$ the 256-token training window; thin lines are three training seeds, thick lines their means. The ordering reverses when the runtime range is retargeted. (d) Crossing factor-four tables derived from the training allocations at 1K: each frozen weight set favors its corresponding table (two seeds, 32 anchors per seed).

Contributions.

- **Identifying allocation effects.** We separate sampled support from interior exponent placement and isolate its effect through paired training, multiple allocation families, and fixed-support frozen interventions.
- **Connecting positional geometry and learning.** We characterize full sine-cosine overlap, its effective-rank identity and slow-frequency limit, and use range retargeting, crossed evaluation, and slot interventions to establish how learned weights use an allocation.
- **Constructing useful allocations.** We develop analytic learning-time and native-relative deployment constructions, with gains in language modeling, complete generation, and natural QA across their respective matched protocols.

2 THE ALLOCATION COORDINATE

Let $K = d_{\text{rot}}/2 \geq 2$ be the number of rotary pairs. Standard RoPE assigns

$$\omega_k = b^{-\phi_k}, \quad \phi_k = \frac{k}{K}, \quad k = 0, \dots, K-1, \quad (1)$$

with base $b > 1$. To separate range from placement, write any strictly ordered positive table $\omega_0 > \dots > \omega_{K-1} > 0$ as

$$x_k = -\log \omega_k = a + Rz_k, \quad z_k = \frac{x_k - x_0}{x_{K-1} - x_0}, \quad z_0 = 0, \quad z_{K-1} = 1. \quad (2)$$

Here (a, R) fixes the sampled log-frequency interval, and the $K-2$ interior values of z allocate the pairs within it. Every geometric table has $z_k = k/(K-1)$, irrespective of base. A base change therefore moves the sampled range while retaining uniform normalized spacing. Changing z at fixed (a, R, K) redistributes the interior frequencies.

This ordered table is installed in labeled Q/K coordinate blocks. A permutation π of the same frequency multiset changes its *slot assignment*; learned projections determine how each assigned frequency contributes to attention. We use these distinctions to design separate interventions on allocation, runtime range, and assignment.

Continuous densities give a convenient way to construct allocations through quantiles. The installed tensor also depends on the finite grid: midpoint nodes, inclusive endpoints, and affine endpoint anchoring yield different tables. Our identification experiments anchor quantiles to the baseline’s actual sampled endpoints. Model applications state their grid convention alongside the comparison. This makes the mathematical coordinate in (2) directly usable as an experimental control.

3 HOW ALLOCATION ORGANIZES THE POSITIONAL BASIS

We first establish that interior placement has a measurable learning effect, then examine the positional functions whose placement is changing.

3.1 A PAIRED INTERVENTION AT FIXED SUPPORT

We train paired 151.9M models with $K = 32$ rotary pairs at length 256. Within each of three seeds, architecture, initialization, token order, optimizer, schedule, and the 499,974,144-token budget are identical. The geometric arm uses the FMROPE training-base rule $b = L_{\text{train}}$ (Oka et al., 2026b); the other anchors Cosh quantiles with $\tau = 4$ to the same endpoints (construction in §5.1). Only the 30 interior exponents differ.

With the training table retained at evaluation, mean Cosh-minus-geometric tail NLL is $+0.026$ at the training length and $-0.281 / -0.176 / -0.146$ at 512/1024/2048 tokens. All three training seeds improve at every extended length (Fig. 1b; full contrasts in Appendix B.1). Thus internal placement can shift the learned length response while frequency endpoints remain unchanged. To see what this intervention changes in the model’s input to attention, we next examine the basis supplied by a rotary pair.

3.2 COMPLETE ROTARY PAIRS AND POSITIONAL OVERLAP

For a fixed content pair, one rotary channel contributes $C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to an attention logit, where Δ is relative token separation. Both coefficients depend on content; its positional object is the full function space

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}. \quad (3)$$

Comparing only cosine curves would privilege one content phase. Instead, let $x_\omega(\Delta) = [\cos(\omega\Delta), \sin(\omega\Delta)]$ and define $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$, $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ under a declared separation measure. For nonsingular self-Grams, the singular values of $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ are the canonical correlations of the two spaces. Their mean square,

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1], \quad (4)$$

measures shared positional directions and is invariant to invertible basis changes within either pair. Let Γ be the Gram matrix of the concatenated block-whitened pairs, with diagonal blocks I_2 . The average pairwise overlap $\bar{c}$ gives the exact Rényi-2 effective-rank identity

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

This connects pairwise overlap to the number of effective directions supplied by the entire positional design matrix: increasing overlap concentrates the same $2K$ coordinates into fewer directions.

3.3 FINITE-WINDOW CONCENTRATION OF SLOW DIRECTIONS

Proposition 1 (Shared slow-frequency subspace). *Under uniform measure on $[0, L]$, with $t = \Delta/L$, as $x = \omega L \rightarrow 0$, $\cos(xt) \rightarrow 1$ and $\sin(xt)/x \rightarrow t$, so V_ω converges to $\text{span}\{1, \Delta\}$. For $\epsilon = \max\{|\omega L|, |\nu L|\} \rightarrow 0$, squared chordal separation satisfies $2 - \|Q_{\omega\nu}\|_F^2 = \mathcal{O}(\epsilon^4)$.*

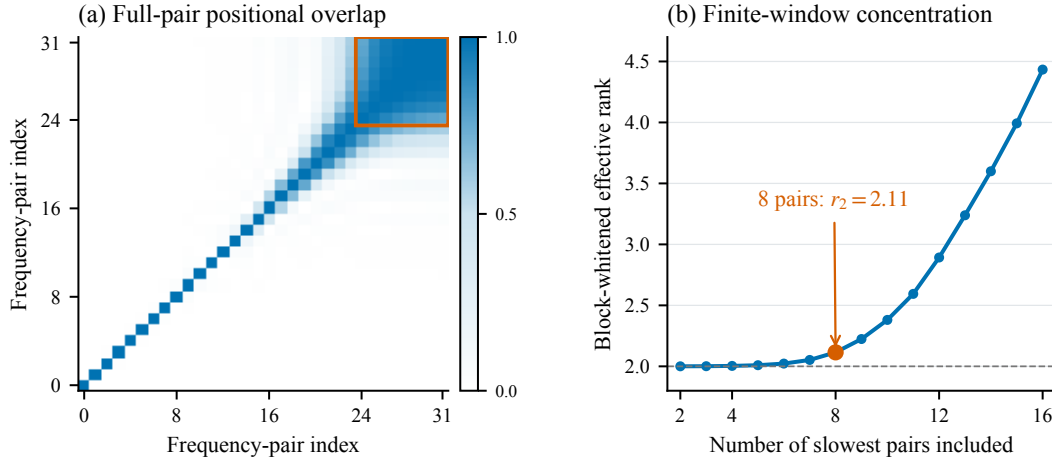


Figure 2: **Log-frequency spacing and positional directions are different objects.** Uniform separation measure on $[0, L]$. (a) Canonical overlap of full sine-cosine pairs on the actual $b = 256$, $K = 32$, $L = 256$ identification grid. The eight slowest pairs share most of their positional space. (b) Effective rank as progressively faster pairs enter that slow subset: eight pairs give 16 coordinates but $r_2 = 2.11$. Values are closed-form Gram calculations checked by independent quadrature.

Distinct slow frequencies therefore approach a shared two-dimensional positional space. On the standard grid $b = 500,000$, $K = 64$, $L = 4096$, the 23 pairs with $\omega L \leq 1$ occupy 46 coordinates but have block-whitened rank 2.00013. The finite-window calculation also reaches the core experiment: at $b = 256$, $K = 32$, $L = 256$, the eight slowest pairs span $\omega L \in [1.19, 4]$ and have rank 2.11 (Fig. 2). Their overlap is substantial even though every frequency has $\omega L > 1$.

The geometry explains why allocating frequencies uniformly in log space need not spread positional directions uniformly over a finite window. It supplies a concrete object for design: where samples are placed changes which scales and directions are represented. Model behavior then depends on the content coefficients learned on this basis. The next interventions examine that learned use directly; complete geometric proofs and cross-length ordering examples appear in Appendix A.1–A.4.

4 HOW A MODEL LEARNS AND USES AN ALLOCATION

A positional basis enters attention through learned Q/K projections. We separate two aspects of its use: the range at which it operates and the association between frequencies and trained coordinates.

4.1 THE LEARNED LENGTH RESPONSE DEPENDS ON RUNTIME RANGE

Return to the paired models of §3.1. Instead of retaining their training support, retarget the log-frequency span to each evaluation length, preserving each allocation’s normalized z . Cosh-minus-geometric NLL becomes $+0.060/+0.227/+0.460$ at 512/1024/2048 tokens: all three seeds now favor the geometric allocation (Fig. 1c). The same trained pair produces opposite orderings under the two range policies. Allocation and range are separately controllable, but their effects interact in the learned model.

4.2 CROSSING WEIGHTS WITH THEIR FREQUENCY TABLES

To isolate learned compatibility, hold each weight set fixed and cross the runtime tables. In a 50M pair, Geo and Cosh achieve matched PPLs of 7.14 and 7.16. Installing the Cosh table into Geo-trained weights raises PPL to 76.20; the reverse swap gives 23.05. A 151.9M crossing repeats the preference across two training seeds using factor-four tables derived from each training allocation (Fig. 1d). The two seed-specific interaction contrasts are 3.400 and 3.251 tail NLL. These crossed comparisons expose the learned association: both allocations support competent models, and each model learns to use its own basis (full crossings in Fig. 8).

4.3 FREQUENCY IDENTITY AND SLOT IDENTITY

The distinction becomes sharper when the frequency multiset itself is preserved. For a block permutation P corresponding to a frequency permutation π ,

$$\mathcal{R}_{\pi\Omega}(\Delta) = P\mathcal{R}_\Omega(\Delta)P^\top, \quad (Pq)^\top \mathcal{R}_{\pi\Omega}(\Delta)(Pk) = q^\top \mathcal{R}_\Omega(\Delta)k. \quad (6)$$

Thus permuting Q/K blocks together with frequencies exactly preserves the attention kernel. With weights fixed, however, permuting only internal frequency assignments raises OLMo tail NLL from 3.10423 to 6.86493 and lowers Qwen 64K task macro from 0.7000 to 0.0000. The model uses an ordered association between coordinate subspaces and frequencies.

For genuinely different frequency multisets, position-independent invertible Q/K maps cannot equate the rotation kernels over an interval containing zero (Theorem 5). Allocation changes the supplied basis; assignment changes its association with learned coordinates. The experimental counterparts establish why a design can be valuable during learning yet require a different installation strategy on a mature checkpoint. We now use this distinction to construct and evaluate allocations in both settings.

5 TURNING ALLOCATION INTO MODEL GAINS

The preceding results identify a useful design freedom and the role of learning in realizing it. We next construct an explicit allocation, test whether its effects extend across shapes, and follow its use through training, continuation, and adaptation.

5.1 AN ANALYTIC ALLOCATION CONSTRUCTION

The overlap analysis motivates spreading a finite budget across useful scales while controlling concentration toward the slow end. We express this geometric motivation as a tractable density-design prior, using a density over normalized exponent space $[0, 1]$. Let $S_\rho(t) = \int_t^1 \rho(\phi) d\phi$, and consider the convex design criterion

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \int_0^1 S_\rho(t)^2 dt, \quad \text{s.t. } \int_0^1 \rho = 1, \rho \geq 0. \quad (7)$$

The first term penalizes concentration; the second penalizes cumulative mass beyond slow-end thresholds. Increasing the concentration penalty favors uniform density; increasing the cumulative-tail penalty moves mass toward smaller exponents. These two terms make the design preference explicit and give a tractable family whose model effects can be measured.

Theorem 2 (Cosh allocation). *For $\alpha > 0$ and $\beta \geq 0$, the unique minimizer of (7) over positive $\rho \in C^2([0, 1])$ with unit integral is*

$$\rho_\tau(\phi) = \frac{\tau \cosh[\tau(1 - \phi)]}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}, \quad (8)$$

recovering uniform density as $\tau \rightarrow 0$. Inverting the CDF yields the explicit quantile map

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k + 1/2}{K}. \quad (9)$$

The density uniquely minimizes (7); EVQ-COSH converts it into a finite table through (9). For $\tau > 0$, ρ_τ is decreasing, so its inverse CDF Q_τ is convex. For equally spaced nodes u_k , affine endpoint anchoring therefore gives

$$z_{\tau,k} = \frac{Q_\tau(u_k) - Q_\tau(u_0)}{Q_\tau(u_{K-1}) - Q_\tau(u_0)} \leq \frac{k}{K-1}, \quad \omega_{\tau,k} \geq \omega_{\text{Geo},k} \quad (10)$$

at fixed (a, R) : convexity places each quantile below the chord between the extreme nodes. Interior pairs move toward faster frequencies while endpoints remain fixed (Fig. 1a). Once τ , grid, and endpoint anchoring are specified, installation evaluates the table once without changing the rotary operator or trainable parameter count.

The strength τ sets this trade-off. We use declared reference settings and compare neighboring strengths and a second analytic family below; Appendix A.11 gives the operating rules and their empirical basis.

Table 1: **Learning-time allocation improves long-context modeling.** Mean PPL within each matched protocol; the endpoint column identifies the frequency-table intervention. The 432M run uses midpoint Cosh quantiles; the 750M continuation anchors inclusive quantiles to the inherited endpoints. The MLA blend is the shared wavelength-dependent evaluation rule in Appendix E.1 ($s = 4$).

Model / setting	Seeds	Train	Endpoints fixed	Evaluate	Geo	Cosh
432M MLA	3	8K	No	8K	35.4	35.8
				16K	138.8	95.6
+ wavelength blend	3	8K	No	16K	117.9	71.1
750M continuation	1	4K	Yes	4K	22.0	22.3
				16K	45.1	24.4

5.2 EVIDENCE ACROSS ALLOCATION FAMILIES

A 50.9M factorial compares geometric allocation, three Cosh strengths, and a deformation-matched exponential family across two bases, two training lengths, three head dimensions, and three paired seeds. Each configuration retains identical sampled endpoints. Reference Cosh improves weighted OOD NLL in 7/12 configurations; the preassigned $1.25\times$ strength improves 10/12, and matched exponential improves 9/12. The reference-Cosh-minus-exponential aggregate is $+0.00074$ NLL with 95% interval $[-0.006, 0.008]$ (Table 7). Multiple analytic shapes therefore realize useful changes along the allocation coordinate. The factorial broadens the single controlled intervention into a design space.

5.3 LEARNING WITH THE CONSTRUCTED BASIS

Allocating scarce rotary channels. In a 432M MLA-style model, only 16 pairs carry explicit rotation (DeepSeek-AI, 2024a;b). Across three seeds, training on 500M tokens at length 8192 with the Cosh table reduces 16K PPL from 138.8 to 95.6, while 8K PPL is 35.4/35.8 for Geo/Cosh (Table 1). Applying the same wavelength-blend rule to both tables gives 117.9/71.1 at 16K; the combined Cosh configuration retains the lowest mean PPL at 24K and 32K. Allocation has practical value when a small explicit rotary budget must support longer contexts.

Learning a new allocation from a shared checkpoint. A 750M pair continues full-parameter training from the same 2K Geo checkpoint for 500M tokens at length 4096. At 16K, PPL falls from 45.1 to 24.4; at 4K it is 22.0/22.3. On 40 associated 8K passkey cases, fixed-length greedy answer-token exact match rises from 0/40 to 31/40, while both teacher-forced retrieval probes remain saturated. Continued learning can therefore turn the allocation change into an improvement in generated answers as well as text likelihood. The single-seed protocol includes 3% passkey and 10% downstream training mix (Appendix D.4).

From distant-source use to complete answers. A matched 300-step Llama-3-8B LoRA pair applies the same 8K adaptation to native-endpoint Geo and midpoint Cosh tables (Hu et al., 2022; Grattafiori et al., 2024). Native/Cosh PPL is 6.82/10.07 at 8K, 108.96/24.07 at 16K, and 991.48/127.91 at 32K. Removing remote gold-block access raises answer NLL by 1.5055 for Cosh versus -0.0095 for Native. Source use improves, while 16K exact retrieval remains zero and the registered QA comparison favors Native (Appendix F.3).

The next question is whether allocation can support successful task readout when continuation explicitly trains it. On OLMo-1.485B, matched 100-step query-gap and 32-step answer-plus-EOS continuations give Native/Cosh complete-string-plus-EOS accuracy of 95%/100% at 4K, 18%/98% at 8K, and 0%/60% at 16K, with 100 prompts per length. These runs use at most 4K physical tokens and target-range position IDs. A separate selective-Q/K continuation improves held-out 2Wiki F1 at 8/16K from 0.07/0% to 21.48/8.57% (200 rows/length). The numeric and QA studies have distinct adapters and supervision; both demonstrate usable long-range outputs under matched allocation comparisons (Appendix F.1).

Further evidence. The same design question extends to 454M three-seed range composition (16K PPL 157.7 $\rightarrow$ 107.5), 1.485B training, video DiT, and a five-configuration GQA/MLA compression study. Their full results and protocols appear in Apps. D.3, F.1, and D.8. Together with the paired identification and factorial, these studies show that allocation is usable through several forms of learning. We next examine how to exploit it when the weights are already fixed.

6 REALLOCATING FREQUENCIES IN MATURE MODELS

The learned associations in §4 make a checkpoint’s native table the natural reference for frozen deployment. We first isolate an allocation effect at fixed support, then construct an intermediate-band adjustment and evaluate its practical use.

6.1 THE ALLOCATION EFFECT SURVIVES WITH FROZEN WEIGHTS

On OLMo-2-1B and Qwen-2.5-1.5B, fix the checkpoint, both sampled endpoints, 64 pairs, attention amplitude, input rows, and decoder. The endpoints span the native fast frequency to one-quarter of the native slow frequency. On OLMo’s nine held-out RULER tasks, uniform normalized exponents achieve 0.56%, a coarse ramp 61.04%, and a residual-guided profile 60.47% (Fig. 3a). Qwen’s four-task development comparison gives 57.75/64.00/66.50%. Thus the allocation effect extends from paired learning to existing checkpoints and task-level generation. The coarse control also shows that much of the effect is captured by the overall profile (Appendix F.2).

Writing $\omega'_k = \omega_k 4^{-m_k}$, a finer control fixes the band, terminal shift, and total displacement $\sum_k m_k = 42$, while changing the interior profile. On the 350-row OLMo development panel, the two profiles score 43.67/54.40%; a separate 16-document instrument gives NLL 2.9417/2.8309. Interior structure therefore carries information beyond the total displacement (Appendix G.7).

6.2 CONSTRUCTING AN ADJUSTMENT RELATIVE TO THE NATIVE TABLE

For native frequencies ω_k^N and installed frequencies ω'_k , define

$$d_k = \log \frac{\omega_k^N}{\omega'_k}, \quad \omega'_k = \omega_k^N e^{-d_k}, \quad \phi'_k - \phi_k^N = \frac{d_k}{\log b}. \quad (11)$$

Frequency blending, log-frequency shifts, and mixed-radix conversion all admit this coordinate (Appendix G). Here s is the configured extension factor; at $s = 1$ the scaled profiles recover the native frequencies. Tables are installed before prefill and retained through decoding. The amplitude $g = 1 + c \ln s$ scales both cosine and sine and therefore contributes g^2 to attention logits. Each frequency comparison fixes g ; joint table–amplitude configurations are reported separately.

MrRoPE-Pro allocates a cumulative shift across an intermediate band (Tian et al., 2026). With $N = h - l$ and $q = \text{clip}(k - l, 0, N)$, its profile and our boundary-matched (BM) construction are

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (12)$$

BM minimizes quadratic difference energy in the radix increments, matching smoothly to the unmodified outer bands. Both profiles retain the band boundaries and terminal extension; their interior cumulative shifts differ (Appendix G.3). This is a deployment construction of the same interior design freedom.

6.3 NATURAL TASKS AND PLACEMENT ON FINITE GRIDS

On the matched OLMo six-task panel, BM(4) reaches 51.32% at 16K; MrRoPE-Uni(4), official YaRN(4), and MrRoPE-Pro(4) score 32.12%, 6.94%, and 2.78% (Appendix G.8). Across five natural-QA tasks, it increases task-equal F1 from 21.62% to 25.44% (95% interval for the difference [1.32, 6.29] points); every task mean improves (Fig. 3b). The complete three-model comparison retains the small Qwen panels, where 128K means favor MrRoPE-Pro: 78.13/70.83% on 3B and 84.44/71.11% on 7B (Table 18).

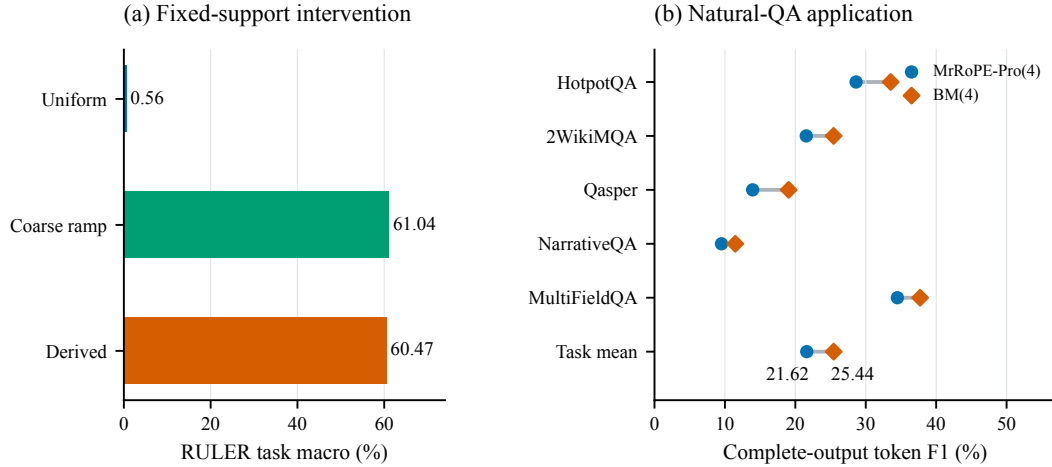


Figure 3: **From a controlled allocation effect to natural-question-answering gains.** (a) Frozen OLMo 16K RULER on nine held-out tasks, 20 rows/task; sampled endpoints, amplitude, weights and inputs are shared. (b) A separate matched $s = 4$ BM/MrRoPE-Pro comparison on five Long-Bench tasks: complete-output token F1 on 631 untruncated inputs longer than 4K. Each panel answers a different question: allocation’s effect at fixed support, and the usefulness of a concrete redistribution.

Useful profiles can also have compact descriptions. A two-parameter clipped-affine law fitted to an OLMo 64-point profile and transported without Qwen refitting achieves 67.75/57.25% at Qwen 64/128K, compared with 67.25/54.50% for the full transported profile. It preserves long-end behavior with a Qwen 32K cost ($82.00 \rightarrow 71.25\%$ relative to Native; Appendix G.6). This connects a successful finite allocation to a simpler transferable construction.

Placement is itself a finite-allocation choice. On Gemma-1.1-2B with 128 pairs (Gemma Team et al., 2024), normalized-index interpolation of a reference displacement profile achieves 79.00% at 16K versus 72.81% for direct-gap placement, a 6.19-point gain ([2.81, 9.63]). A Qwen-0.5B static table–amplitude configuration improves 64K RULER-13 by 6.09 points over the evaluated YaRN(2) configuration while retaining near-equal 32K scores. Apps. G.1–G.2 give the paired placement and joint-configuration comparisons; Table 17 separates table and amplitude contrasts on the small Qwen-3B panel. These results extend allocation design from continuous profiles to their installation on real model grids.

7 RELATED WORK

Range and frequency allocation. RoPE’s rotation operator (Su et al., 2024) admits many frequency choices. Scaling-law analyses study how geometric range affects length generalization (Liu et al., 2024; Xu et al., 2024). FMRoPE relates the learned band to base and training length (Oka et al., 2026b). In that family normalized exponents remain equally spaced; our fixed-range experiments isolate interior allocation. Position interpolation scales positions uniformly (Chen et al., 2023a), while YaRN blends frequencies across bands (Peng et al., 2024). LongRoPE searches nonuniform interpolation factors and uses progressive extension (Ding et al., 2024); LongRoPE2 uses needle-driven perplexity for search and mixed-window training to preserve short-context performance (Shang et al., 2025). Our paired training isolates allocation at matched endpoints, and our constructions specify analytic tables. MrRoPE parameterizes intermediate-band changes through mixed-radix conversion (Tian et al., 2026). Equation (54) casts these modifications as exponent displacements; our work separates their interior profile from sampled range.

Learning and using frequency bands. LeRoPE learns shared per-channel scales and tests training with a separately learned fixed table, while AdaRoPE studies head-specific frequencies and attention scaling (Karypis et al., 2026; Wang et al., 2026). These studies establish the value of flexible frequencies. Our exact-endpoint interventions identify the contribution of normalized inte-

rior placement, and crossed evaluation connects that placement to weights learned under each table. Frequency-use studies connect rotary coordinates to positional versus symbolic attention, retrieval, and data-dependent scales (Barbero et al., 2025; Urrutia et al., 2026; Chiang & Yogatama, 2025; Oka et al., 2026a; Wu et al., 2026). Urrutia et al. (2026) study symbolic and positional attention through model-dependent measurements. Our shared slow-frequency subspace concerns positional functions, leaving their learned content roles to such measurements. DoPE uses truncated matrix entropy to selectively modify positional encodings (Xiong et al., 2025). Our canonical correlations instead characterize supplied sine-cosine subspaces as exponents vary.

Möbius RoPE uses an anti-periodic frequency ladder and tests frozen table swaps (Bae, 2026). Its matched boundary-condition controls and our fixed-endpoint allocation controls address complementary aspects of frequency design.

Changing the positional operator. FoPE introduces Fourier components (Hua et al., 2025), while HoPE and clipped encodings alter slow rotations (Chen et al., 2025; Li et al., 2026a). Other variants explore group actions (Zhang et al., 2026), input-dependent rotations (Movahedi et al., 2026), contextual token positions (Li et al., 2026b), and compression preservation (Huang et al., 2026). Long-context token confusion analyses further motivate studying frequency choices and learned representations (Gu et al., 2026; Du et al., 2026). Our constructions retain the standard rotary map, while the controlled studies examine how its finite frequency table and learned coordinate associations affect behavior.

8 DISCUSSION AND CONCLUSION

Interior exponent allocation connects three levels of RoPE design. It organizes the positional functions supplied by a finite table; training associates those functions with learned projections; explicit constructions use that freedom to improve model behavior. Fixed-support training and frozen interventions establish the effect, full-pair geometry identifies the positional object, and range retargeting and crossings reveal how the learned model uses it. The resulting designs have practical value across language modeling, complete generation, and natural question answering.

This view suggests a concrete design practice: specify the sampled range and interior placement separately, evaluate learning-time allocations with paired training, and install mature-model adjustments relative to the native table while controlling amplitude. Analytic profiles provide reproducible starting points; multi-shape comparisons and task-level measurements determine their value in a given setting.

Scope of the evidence. The primary training identification uses 151.9M models at a 256-token window, complemented by a 50.9M configuration factorial and fixed-support mature-model tasks. Larger learning studies supply evidence under their own grid and adaptation protocols. Exact geometry describes the positional basis, while downstream effects are measured by the model experiments; the Cosh optimum refers to its stated density criterion. Full length curves, task outcomes, and exploratory profile diagnostics are retained in the appendices. The collective result is a useful additional design coordinate for RoPE: how a finite rotary budget is allocated within its frequency range.

ETHICS STATEMENT

This work introduces no new dataset, deployed decision system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences. Improved long-context processing does not by itself establish factual reliability, privacy protection, or safe deployment; claims in this work concern the reported positional, language-modeling, and task endpoints.

REPRODUCIBILITY STATEMENT

The appendices specify the installed frequency tables, exponent grids, attention amplitudes, training and adaptation recipes, datasets, and evaluation procedures. The fixed-range experiment reports three paired training seeds; the larger-model and frozen-model tables identify their comparison units and sample counts. The accompanying source archive contains the active TeX, bibliography, plotted figures, numerical summaries, a SHA256 manifest, and standalone verification of the explicit Gram-matrix examples. Experimental checkpoints and public datasets are documented separately, together with the preprocessing required for selected reruns.

AI USE STATEMENT

We used generative AI tools to assist with mathematical discussion, code development, literature organization, and manuscript editing. Experimental results are drawn from recorded model runs and numerical analyses. The authors are responsible for the correctness and originality of the submitted text, mathematical arguments, experimental claims, and accompanying code.

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GUIDE TO THE SUPPLEMENTARY MATERIAL

The supplementary material follows the paper’s argument, expanding each connection between geometry, learning, and practical allocation. Table 2 provides entry points to the complete results and proofs.

Table 2: **Where each part of the argument is developed.**

Question	Supporting material
What does the allocation supply?	Full-pair geometry, finite-window calculations, and static ordering examples (Apps. A.1–A.4).
What is learned from it?	Paired training and range retargeting (App. B.1); crossed weights/tables (App. B.2, Fig. 8); assignment and kernel equivalence (App. A.5).
How can allocations be constructed?	Cosh derivation and finite quantization (App. A); multi-shape factorial (App. B.3); learned-frequency comparator (App. C).
What happens during learning?	Training protocols and 750M continuation (App. D); full MLA curves (App. E.1); 1.485B, selective-Q/K, and 8B adaptation (App. F); video DiT (App. D.6).
How does a mature model respond?	Fixed-support controls (App. F.2); static profiles, placement, BM, and natural QA (App. G); measured profile diagnostics (App. G.11).

A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (13)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. Thus $c_{\omega\nu}$ is an intrinsic measure of overlap between the two positional subspaces.

Theorem 3 (Spectral budget identity). *For the block-whitened Gram matrix Γ with diagonal blocks I_2 , $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K - 1)\bar{c}]$. Hence its Rényi-2 effective rank is given by (5).*

Proof of Theorem 3. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K - 1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (5). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 1. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (14)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (15)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 . Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (16)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (17)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (18)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1}M) + O(\epsilon^6) = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6), \quad (19)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \rightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{\overline{2\cos(\omega\Delta)}}{\omega^2} \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (20)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. The separation weights p therefore determine which polynomial directions survive centering. $\square$

Directional overlap and feature scale. For the original, unwhitened pair on $[0, L]$, put $x = \omega L$ and $\text{sinc } x = \sin x/x$. The self-Gram eigenvalues are

$$\lambda_\pm(S_\omega) = \frac{1 \pm |\text{sinc } x|}{2}, \quad \lambda_-(S_\omega) \sim \frac{x^2}{12}, \quad \kappa(S_\omega) \sim \frac{12}{x^2} \quad (x \rightarrow 0).$$

Thus the slow pair has a poorly scaled second direction before normalization. Block whitening removes this within-pair scaling and measures overlap between positional directions. Raw feature energy, coefficient magnitude, and numerical precision determine how strongly those directions can contribute to a model.

The standard $u_k = k/K$ grid gives 23 slow pairs in Figure 4; the endpoint-inclusive grid used in Figure 5 gives 24 at $K = 64$. Both figures use the uniform separation measure.

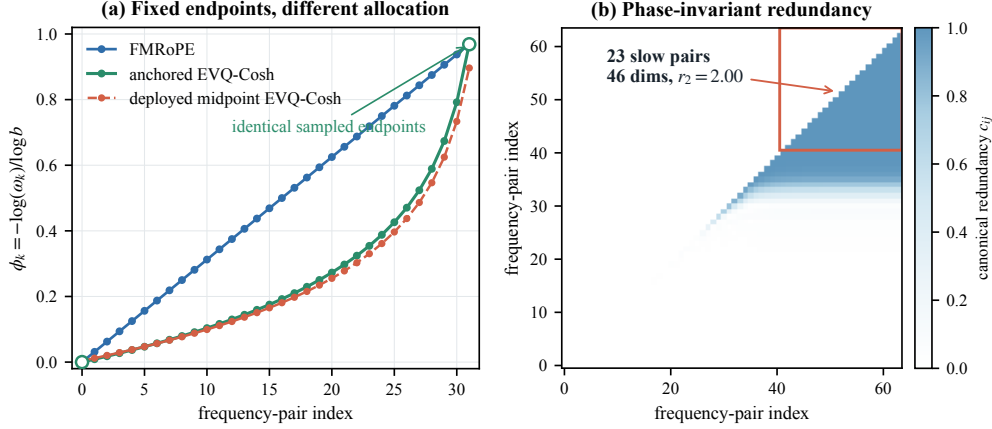


Figure 4: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

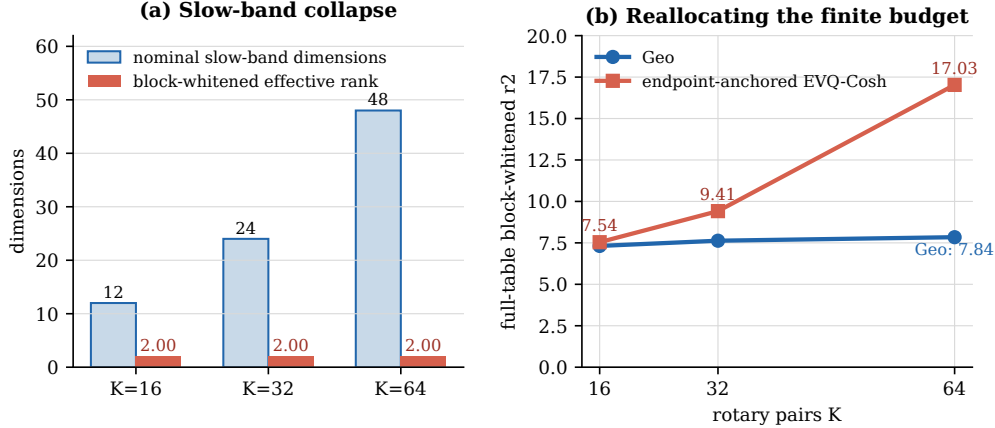


Figure 5: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows.

A.2 FINITE-WINDOW OVERLAP IN THE IDENTIFICATION EXPERIMENT

The slow-limit proposition has a finite-window counterpart that can be calculated directly on the identification grid. Set $b = 256$, $K = 32$, $L = 256$, and retain the n slowest frequencies of $\omega_j = b^{-j/K}$. Their dimensionless frequencies are $\omega_j L = 1.1892, 1.4142, 1.6818, 2, \dots$ in slow-to-fast order. The smallest is already greater than one, so we evaluate the full cross-Gram in (13) without a small-phase approximation.

For $n = 8$, the largest two Gram eigenvalues are 7.9681 and 7.5752, accounting for 97.1% of the trace; the third is 0.4246. As faster pairs enter, additional positional directions acquire mass. This is the finite-window overlap represented in Fig. 2. The accompanying `figs/make_story_figures.py` checks the $n = 8$ closed-form calculation against independent 256-node Gauss-Legendre quadrature and also recovers $r_2 = 2.00013$ for the 23 slow pairs of the $b = 500,000$,

Table 3: **Finite-window positional concentration on the paired-training grid.** The separation measure is uniform on $[0, L]$. The nominal coordinate count is $2n$; r_2 is the effective rank of the block-whitened Gram.

Slowest pairs n	Largest ωL	Coordinates	r_2
2	1.41	4	2.00030
4	2.00	8	2.00361
6	2.83	12	2.02263
8	4.00	16	2.11474
12	8.00	24	2.89256
16	16.00	32	4.43339

$K = 64$, $L = 4096$ grid. The machine-readable calculation is in `figs/finite_window_geometry.json`.

A.3 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction that also identifies the complete parity class omitted by the usual $2\pi/L$ Fourier description.

Proposition 4 (Parity-lattice orthogonality and recurrence). *Let $a_1, \dots, a_K$ be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (21)$$

Under $\Delta \sim \text{Unif}[0, L]$, the K full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is $\Gamma = I_{2K}$ and $r_2(\Gamma) = 2K$. If support is free but restricted to $0 < \omega \leq 1$, the largest same-parity class in this construction has

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (22)$$

pairs. The resulting feature map is recurrent: for even a_k , $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$; for odd a_k , $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$ and hence $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$.

Proof. For $i \neq j$, same parity makes both $a_i - a_j$ and $a_i + a_j$ nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (23)$$

so every entry of the cross-Gram in (13) vanishes: $a((\omega_i \mp \omega_j)L) = 0$ and $b_\star((\omega_i \mp \omega_j)L) = 0$. Likewise $2\omega_i L = 2\pi a_i$ gives $S_{\omega_i} = \frac{1}{2}I_2$. Thus the unwhitened Gram is $\frac{1}{2}I_{2K}$ and block whitening gives $\Gamma = I_{2K}$. Among the integers $1, \dots, \lfloor L/\pi \rfloor$, the larger parity class has $\lceil \lfloor L/\pi \rfloor / 2 \rceil$ elements, proving (22) for this construction. Finally, $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k\Delta}$; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after $2L$. $\square$

Proposition 4 couples maximal rank on $[0, L]$ with recurrence beyond that interval. Its orthogonality also depends on the separation measure: weighting the same lattice by the triangular causal histogram introduces off-diagonal Gram terms. Thus interval coverage and pairwise rank capture different features of a positional basis.

A.4 TWO STATIC ORDERING COUNTEREXAMPLES

Two explicit examples show how phase coupling and interval length change the ordering of frequency tables.

Cosine-only collision can choose the lower-rank table. Take $K = 4$, $L = 8\pi$, and the two descending frequency tables

$$\Omega_A = (1, 5/8, 3/8, 1/4), \quad \Omega_B = (1, 6.01/8, 4.01/8, 1/4).$$

They share both endpoints. Define $C_{\cos}$ as the mean squared normalized cosine-cosine overlap over distinct pairs. For Ω_A , every cosine is an integer harmonic on $[0, L]$, so $C_{\cos}(A) = 0$ exactly. Mixed harmonic parities nevertheless produce nonzero sine-cosine cross terms. For Ω_B , the small perturbation of an even-harmonic table gives

$$C_{\cos}(A) = 0 < 1.38725 \times 10^{-5} = C_{\cos}(B), \quad (24)$$

$$r_2(A) = 6.29478 < 7.99971 = r_2(B). \quad (25)$$

Thus a strict cosine-only preference can have the opposite full-subspace rank ordering. Both values follow directly from Eq. (13).

Full-subspace collision can reverse across lengths. Now use $K = 4$, $L = 2\pi$, and the fixed tables

$$\Omega_A = (1, 0.99, 0.02, 0.01), \quad \Omega_B = (1, 2/3, 1/3, 0.01).$$

Again the endpoints agree. Writing C_M for mean phase-invariant collision under the uniform measure on $[0, M]$, Eq. (13) gives

$$C_L(A) = 0.538996 < 0.631721 = C_L(B), \quad (26)$$

$$C_{2L}(A) = 0.384445 > 0.204995 = C_{2L}(B), \quad (27)$$

$$C_{4L}(A) = 0.345188 > 0.033506 = C_{4L}(B). \quad (28)$$

The close pairs in A retain high overlap as the interval grows, while the more evenly spread frequencies in B become easier to distinguish. The example makes the interval dependence of the geometry explicit.

All frequency entries and measures are specified above. The accompanying `figs/verify_explicit_geometry.py` evaluates the closed-form Gram and checks it against independent 128-point Gauss-Legendre quadrature; the largest discrepancy in the displayed metrics is below 4×10^{-14} . These finite basis examples complement the trained allocation comparisons and the weight-table crossings.

A.5 POST-HOC TRANSPLANT OBSTRUCTION

Table 4: Weights co-adapt to their training-time frequency table. Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head-query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

Theorem 5 (Post-hoc transplant obstruction). *Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta) B = \mathcal{R}_\Omega(\Delta)$ on an open interval containing zero, then Ω' and Ω have the same frequency multiset up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

Proof of Theorem 5. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum.

The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace, including across its original two-dimensional blocks. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

Approximate compatibility within a geometric family. The exact kernel result leaves room for approximate recovery of a model’s behavior. In the historical 50M crossing, three geometric controls were derived from the Cosh table without fitting model loss: matching its slow endpoint ($b = 331.6\text{K}$), matching its log span (377.7K), and minimizing log-frequency RMS within the geometric family (8.06K). On Cosh-trained weights these give PPL 21.69, 22.14, and 9.63, compared with 23.05 for the original Geo table and 7.16 for the training Cosh table. A coarse geometric approximation can therefore recover substantial frozen compatibility. On Geo-trained weights the respective PPLs are 7.30, 7.16, and 45.63 (original Geo: 7.14). These historical tables also change support, so the fixed-support training and mature-model controls supply the separate interior-allocation estimates.

A.6 SLOT ASSIGNMENT AND CONDITIONAL PREDICTION ERROR

Let $\mathcal{R}(\Omega; M, D)$ denote the risk of a fixed model M on a fixed evaluation set D when frequencies Ω are assigned to labeled rotary slots. This domain includes unsorted assignments. It differs from the strictly ordered allocation domain in §2, where the full multiset recovers the sorted table.

Proposition 6 (Prediction error on an observed permutation pair). *Suppose a pair $\Omega, \pi\Omega$ has risk gap $\Delta_R = |\mathcal{R}(\Omega; M, D) - \mathcal{R}(\pi\Omega; M, D)| > 0$. For every permutation-invariant feature map Φ and every scalar predictor h ,*

$$\max_{\Xi \in \{\Omega, \pi\Omega\}} |h(\Phi(\Xi)) - \mathcal{R}(\Xi; M, D)| \geq \Delta_R/2.$$

Proof. Invariance gives the same prediction $c = h(\Phi(\Omega))$ for both assignments. The triangle inequality bounds Δ_R by the sum of the two prediction errors, and hence by twice their maximum. $\square$

The risk-gap premise is empirical. It does not follow from unequal projection norms or from Theorem 5. Synchronously transforming the Q/K blocks compensates the permutation exactly by (6); changing the table while freezing those projections measures a different intervention. In the measured pair, OLMo-1B PG-19 tail NLL changes from 3.10423 to 6.86493, and Qwen-2.5-1.5B 64K core-four macro changes from 0.7000 to 0.0000.

Discrete orbit-count sensitivity. In the archived scale-orbit construction at factor four, one-ULP float32 jitter changes the exact orbit count from 6 to 64, while the 16K tail-NLL difference (jitter minus exact) is -0.000013 (paired interval $[-0.003682, 0.003715]$) and the 16K core-four score remains zero. This rejects the interpretation that a large increase in this exact algebraic count necessarily produces an appreciable loss change on the tested finite horizon. It does not rule out a predictor that maps different counts to the same risk. Discrete-period design, as studied by Resonance RoPE (Wang et al., 2024), should be distinguished from this exact-ratio sensitivity test.

Finite coordinate perturbations and continuity. The clipped-affine C^2 approximation has movement MAE 0.001223 and RMSE 0.006118 relative to its reference profile. The evaluated OLMo retention score is 0.870971, below the 0.8750 floor. This finite example shows that this approximation tolerance did not certify retention. It establishes neither a Hessian eigenvector nor failure arbitrarily close to the reference. If fixed-input NLL is continuous and a reference has positive margin below an NLL gate, a sufficiently small neighborhood passes that gate by continuity.

A.7 FINITE-PHASE OPERATOR CHANGES

For $\delta = (\omega' - \omega)\Delta$, the second-order cosine remainder is bounded by $|\delta|^3/6$. The series converges for all δ , but a low-order approximation need not be accurate at large phase increments. An exact

finite-phase operator bound is

$$\|\mathcal{R}_{\Omega'}(\Delta) - \mathcal{R}_{\Omega}(\Delta)\|_2 = 2 \max_j \left| \sin \frac{(\omega'_j - \omega_j)\Delta}{2} \right| \leq \max_j \min\{2, |\omega'_j - \omega_j| |\Delta|\}. \quad (29)$$

A.8 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 2 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (30)$$

the allocation functional is (7), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (31)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (32)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (33)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (34)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (33) and (34) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (35)$$

which is main-text Theorem 2. Mass normalization enters the boundary condition through $g'(0) = \int_0^1 \rho = 1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1) = \tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (35).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (36)$$

The strictly convex functional on $L^2([0, 1])$ therefore has the displayed stationary density as its unique constrained minimizer.

The Cosh solution has an analytically invertible CDF, so its quantiles turn the continuous allocation directly into a finite frequency table.

A.9 SINGLE-CROSSING BUDGET SHIFT

Lemma 7 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (37)$$

Proof of Lemma 7. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (37). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. $\square$

A.10 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 8 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (38)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (39)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (40)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (38). $\square$

An immediate corollary follows.

Corollary 9 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (41)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2 \sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity provides a closed-form evaluation of the allocation energy.

A.11 ALLOCATION STRENGTH AND A REFERENCE RULE

The Cosh family varies with the strength τ . A reference used in the full-RoPE text experiments relates this strength to head width and training length:

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \tfrac{1}{2}) / K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (42)$$

The reference sets $c = 1$ before training. Table 10 records the strength, grid, and setting procedure used in each experiment.

Local scaling calculation. The scaling argument uses four explicit modelling assumptions: (i) the pure-tether Cosh family is varied only through τ ; (ii) full-RoPE MHA has $d_{\text{rot}} = d_{\text{head}}$; (iii) the reference post-softmax distribution is diffuse, $p_0 = 1/L$; and (iv) a fixed trade-off $\lambda > 0$ compares a normalised channel-load stiffness with a specified phase-variance utility, with $Q_1(L, b) > 0$. Under those assumptions and at small τ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45d_{\text{head}}} + O(\tau^6), \quad (43)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (44)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left(\frac{(1-\phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2. \end{aligned} \quad (45)$$

Stationarity of $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ gives the leading small- τ balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (46)$$

This calculation yields the scaling factor $d_{\text{head}}/\sqrt{L}$ with a local coefficient determined by $Q_1(L, b)$ and λ . The experiments evaluate the exact Cosh quantiles at the protocol’s chosen strength.

Reference and neighboring strengths. The staged study contains 99 completed runs: the fixed $c=1$ reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at 0.75 $\times$, 1.25 $\times$, or 1.5 $\times$; the independent exact-range factorial tests 0.75 $\times$, 1.00 $\times$, and 1.25 $\times$, and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction endpoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. The reference and its neighboring strengths give a direct comparison of allocation intensity across the tested configurations.

A.12 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. We compare the measures in Wasserstein distance and approximate the density with the associated quantile-cell histogram ρ_K .

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (47)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (48)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$, $m_\tau = \tau / \sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (49)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (50)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoﬀ, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (51)$$

This expansion quantifies how finite exponent samples approximate a continuous allocation under a smooth distortion. For endpoint-anchored comparisons, $K = 2$ leaves no interior entries and the geometric and Cosh grids coincide.

Large- τ regime. At large τ , $m_\tau = \tau / \sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B CONTROLLED ALLOCATION EXPERIMENTS

B.1 EXACT-RANGE CONTROL AT 151.9M

Paired controls. The two trained tables have identical frequency endpoints. Within each seed pair the arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMRoPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the same normalised shape: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Writing Φ_τ for the inverse CDF in (9), let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R \frac{k}{K-1}\right). \quad (52)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed*: each checkpoint retains its training range. *target-matched*: the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$. The anchors are common end positions selected with Python’s `random.Random(20260723)`: the eligible validation stream is divided into 32 equal-width bins, with one end position drawn per bin and a 2,048-token guard before the next bin. At anchor e , length L uses validation tokens $[e-L, e)$ and scores the final 128 targets $[e-128, e)$. Longer contexts therefore add earlier text around the same target tokens; the longest windows at distinct anchors are disjoint.

Exact range transformation.

$$R(L) = \frac{K-1}{K} \log L, \quad \omega_k^{\text{target}}(L) = \exp[-R(L)z_k].$$

Here $z_k = k/(K-1)$ for FMRoPE and the anchored Cosh shape for its paired arm. The fixed condition uses $L = 256$ at every evaluation length; the target-matched condition substitutes the declared evaluation length. Both conditions use amplitude one.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 5: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218

Across all three seeds, fixed training range favors Cosh at every extended length, whereas target-matched range favors FMRoPE. The paired design thus shows both an interior-allocation effect and a change in preferred allocation when the same trained weights are evaluated under a retargeted range.

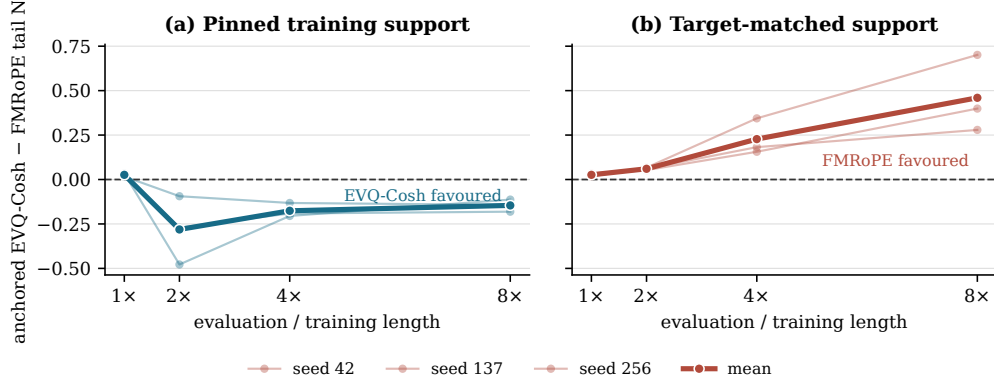


Figure 6: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

B.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of z . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired FMRoPE and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

For each training table Ω , the derived table uses the following label-free residual construction. Set $L_0 = 256$, $\Delta_j = 2j$ for $j = 0, \dots, 127$, and $p_j = (2L_0 - 4j - 1)/(L_0(L_0 + 1)/2)$, the grouped causal-separation weights. Let X_k have columns $\sqrt{p_j} \cos(\omega_k \Delta_j)$ and $\sqrt{p_j} \sin(\omega_k \Delta_j)$, and let X_{-k} collect all other pairs. Define

$$U_k = \frac{\|X_k - X_{-k} X_{-k}^\dagger X_k\|_F^2}{\|X_k\|_F^2}, \quad \bar{U}_k = \frac{U_k - \min_i U_i}{\max_i U_i - \min_i U_i}, \quad w_k = (1 - \bar{U}_k)^2.$$

The pseudoinverse uses relative singular-value cutoff 10^{-10} ; the zero-span convention is $\bar{U}_k = 0$. The frequency blend is $\omega'_k = \omega_k(1 - 3w_k/4)$, cast to FP32 with its endpoints set exactly to $\omega'_0 = \omega_0$ and $\omega'_{K-1} = \omega_{K-1}/4$. Both derived tables use $g = 1 + 0.1 \log 4$ and remain unchanged across the 512- and 1,024-token evaluations.

Table 6: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	FMRoPE-derived table	Anchored-EVQ-COSH-derived table
FMRoPE-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

With $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$, the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit; the 32 anchors are paired observations within a seed. Both seeds exhibit the same preference for the runtime table derived from their training allocation.

Table 7: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times / 4 \times / 8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. B).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	− 0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

B.3 EXACT-RANGE FACTORIAL AT 50.9M

Design. The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M $\times$ training length 256/1,024 $\times$ 4/8/16 heads ($d_{\text{head}}=128/64/32$), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 raw stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the $1.25 \times$ multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at $0.75 \times$, $1.00 \times$ (the formula point), $1.25 \times$, and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- τ follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75 \times$ 6.202858 (494.16); EVQ-COSH $1.00 \times$ 6.202094 (493.78); EVQ-COSH $1.25 \times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5 \times$ arm is better; at formula $\tau=8$ the $0.75 \times$ arm is. The improving direction points toward the intermediate strengths at both tested extremes.

The factorial varies architecture, base, and training length on a repeated WikiText-2 stream (Merity et al., 2017). Its configuration-level contrasts complement the longer FineWeb-Edu training comparison in §B.1.

B.4 FINITE DEPLOYMENT INTERVENTIONS

The permutation and finite- $C2$ results in Appendix A.6 concern fixed-weight interventions. The $C2$ approximation has its largest coordinate residual at slot 19. With zero-based indexing and native $K = 64$ geometric frequencies, that slot makes 13.25 turns in OLMO’s 4096-token window ($b = 500,000$) and 86.30 turns in Qwen’s 32768-token window ($b = 1,000,000$). Neither is the native one-turn boundary. A transition in an installed profile and the one-turn boundary of a native table are distinct quantities.

Three-band profiles retain a fast band, fully scale a slow band, and distribute the shift across an intermediate band. The cutoffs specify a construction; performance is measured with the checkpoint and gain held explicit. On the OLMo six-task confirmation panel, BM improves 16K macro from 2.78% to 51.32% relative to MrRoPE-Pro. On the separate 631-prompt natural-QA long-input stratum it improves task-equal F1 from 21.62% to 25.44%. Detailed model contrasts and scoring protocols are in Appendix G; the exploratory twelve-profile panel is documented in Appendix G.11.

C LEARNED-FREQUENCY COMPARATOR

Table 8: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8K$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). The learned-inverse-frequency arm has one parameter per pair, shared across layers and heads.

Method ($128 \rightarrow 8K$)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 ± 1.3	437.9 ± 12.2	−14.8%
Learned inv-freq ($100\times$ positional LR)	32	183.6	455.3	−11.4%
same, $10\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm learns one inverse frequency per rotary pair, shared across layers and heads. This is the learned-table parameterization also studied by Karypis et al. (2026). Its positional-parameter learning-rate sweep uses $10\times$ and $100\times$ the base rate; the stronger $100\times$ setting gives 455.3 PPL. The learned table and the fixed Cosh allocation both improve on Geo in this protocol.

D EXPERIMENTAL DETAILS

D.1 EXPERIMENT ORGANIZATION

This section gives the training, continuation, and video protocols. Apps. B and F detail the fixed-range and mature-model experiments.

D.2 MODELS AND TRAINING SETTINGS

Table 9 collects the model settings and evaluation units. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 11 lists the training hyperparameters shared across the supporting text experiments.

Within the scratch and continuation comparisons specified in this section, the architecture, optimizer, data order, and training loop are fixed, and the intervention is the initialized inverse-frequency table. The released-checkpoint 1.485B comparison in Appendix F.1 uses separate trainer implementations, as detailed there. The scratch-text experiments use base $b=500K$ unless stated otherwise.

Table 9: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime and intervention	Unit, endpoint, and appendix
Allocation identification	151.9M scratch; fixed support, 30 interiors	3 training seeds; tail NLL; App. B.1
Co-adaptation replication	151.9M frozen crossing; weights $\times$ runtime table	2 training seeds; tail NLL; App. B.2
Shape/configuration breadth	50.9M factorial; fixed support, 5 allocations	12 configs $\times$ 3 seeds; weighted OOD NLL; App. B.3
Scarce rotary budget	432M MLA-style; Geo/EVQ-COSH, $K=16$	3 training seeds; PPL; App. E.1
Full-parameter scale	750M/1.485B; continued/from-init Geo/EVQ-COSH	one comparison each; PPL/retrieval; Apps. D.4/F.1
Frozen allocation	OLMo/Qwen; same support and amplitude	20 rows/task; RULER; App. F.2
Zero-training deployment	OLMo; frozen long table, gain, session route	fixed checkpoint/rows; NLL/RULER/F1; App. F.2
Adaptation/source use	1.485B/8B; matched LoRA/source block	one trained pair; QA/RULER/NLL; Apps. F.1/F.3
Static index profile	Qwen-0.5B; one table across lengths	13 tasks, 20 rows/task/length; App. G.1
Profile placement	Qwen/Gemma; direct gap versus normalized index	4 tasks, 80 rows/task; App. G.2
Intermediate-band shape	OLMo/Qwen; matched BM/MrPro profiles	6 tasks; Table 18; App. G.4
Natural-QA transfer	OLMo; matched static BM/MrPro profiles	778 inputs, 631 long; token F1; App. G.5
Video modeling	129.6M video DiT; temporal Geo/EVQ-COSH	1 training seed; denoising MSE; App. D.6

Table 10: **Allocation protocols, parameter bounds, and float32 tensor SHA-256 digests. Top:** Cosh-family settings across reported experiments. Midpoint means $u_k = (k + 1/2)/K$; standard means $u_k = k/K$; anchoring fixes sampled endpoints after quantile evaluation. **Bottom:** Parameter bounds and float32 tensor hashes for static deployment profiles. MrRoPE-Pro parameters follow discrete turn cutoffs $\beta = 32, \alpha = 1$ with fast cutoff $l = \max\{j \mid \text{turns}_j > 32\}$, slow cutoff $h = \min\{j \mid \text{turns}_j < 1\}$, and transition width $n = h - l$. FullLagP2 evaluates all 32,768 integer lags, eliminating p2 aliasing spikes.

Part A: Cosh-Family Pre-training and Adaptation Settings			
Protocol	τ	Grid	Setting convention
151.9M fixed-range	4	Anchored midpoint	Fixed $64/\sqrt{256}$ reference
50.9M factorial	Varies	Anchored midpoint	Three preassigned multiples of $d_{\text{head}}/\sqrt{L_{\text{train}}}$
50M crossing	2.83	Midpoint	Reported training-table setting
432M MLA	1.414	Midpoint	Pre-specified architecture-specific setting
454M composition	1.5	Midpoint	Reported training recipe
750M continuation	1.5	Anchored inclusive	Fixed continuation recipe; $u_k = k/(K - 1)$
Llama-3-8B	1.414	Midpoint	Fixed adaptation recipe; approximately $128/\sqrt{8192}$
OLMo-2	2	Standard	Fixed $128/\sqrt{4096}$ reference
129.6M Video DiT	1.5	Midpoint	Best tested value in the video sweep
Part B: Static Deployment Profiles, Bounds, and Tensor SHA-256 Digests			
Profile / Architecture	Scale s	Bounds / Lags	Float32 Tensor SHA-256 Digest
MrRoPE-Pro (Qwen2.5)	4	$l=23, h=40, n=17$	33cbe3a4 0994ac2a 79126a14 ce302828 67bd6d49 b74c1ada 4d4cce8a 7e76016f
MrRoPE-Pro (OLMo-2)	4	$l=14, h=32, n=18$	ec9c5497 77f7a94c 73a1cc2d b5f22fef be202a7e 083e454f 2510034f 2cec5dfb
FullLagP2 (Qwen2.5)	4	32,768 integer lags	ecd0c280 a11788e0 c4a869a3 d1628809 64ba3f37 1f589f7e 2f5ae471 d461096b
Boundary-Matched (OLMo-2)	4	$s=4, g=1.1386$	c80626f8 f9280854 577fd1e7 94439300 09a69370 65366adb 8eee5f62 ce7807a6
Native RoPE (OLMo-2)	1	$L_0=4096, b=500K$	9229ec6a 6a7ce5c1 750273bc dbec9789 5a74a920 2722f14b 6ca2328e ce5ee98e

Table 11: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.95, \epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 14 (750M: micro 7, accumulation 2), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

D.3 COMPOSITION WITH A FIXED-INDEX RANGE OPERATOR

The 454.2M decoder has 24 layers, width 1024, 16 heads, $d_{\text{head}} = 64$, and base $b = 500,000$. Geo ($\tau = 0$) and EVQ-COSH ($\tau = 1.5$) use midpoint frequency grids and train from initialization for 100M tokens per arm at length 2048, using FineWeb-Edu with a 10% synthetic-passkey mixture and seeds 42, 123, 7. The same inference-time operator is then applied to both trained tables.

For $k = 0, \dots, K - 1$, define

$$\ell = \lfloor 0.20K \rfloor, \quad h = \lfloor 0.90K \rfloor, \quad t_k = \text{clip}\left(\frac{k - \ell}{h - \ell}, 0, 1\right), \quad r_k = t_k^2(3 - 2t_k).$$

The repository’s fixed-index, YaRN-style smooth-ramp operator installs

$$\mathcal{R}_s(\omega_k) = \frac{\omega_k}{s^{r_k} T(s)^{r_k/2}}, \quad T(s) = 1 + 0.07 \log_2 s. \quad (53)$$

Here $K = 32$ and $s = 8$, giving $(\ell, h, T) = (6, 28, 1.21)$. The temperature factor is folded into the frequency divisor; cosine/sine amplitude stays at $g = 1$. Equation (53) specifies the operator used in this experiment, separately from the official-equation YaRN comparisons in §6. Table 12 reports the three-seed comparison.

Table 12: Matched fixed-ramp composition on the 454M three-seed experiment. $\mathcal{R}_8$ is defined in Eq. (53). PPL uses full-sequence scoring; PK is teacher-forced NLL-gap retrieval. Entries are three-seed means; PK@8K also gives the standard deviation.

Table	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41±5%	57%	51%	161.9	253.2
Geo+ $\mathcal{R}_8$	61±3%	59%	51%	82.9	157.7
EVQ-COSH	53±8%	63%	50%	150.3	229.5
EVQ-COSH+ $\mathcal{R}_8$	100±0%	79%	68%	70.9	107.5

D.4 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, fixed-length autoregressive answer-token exact match is 0% versus 77.5%. The EVQ-COSH continuation uses $\tau = 1.5$, $K = 32$, and the endpoint-inclusive quantiles $u_k = k/(K - 1)$. Writing q_τ for the inverse CDF in Eq. (9), the installed frequencies are

$$\omega_k = \exp\left[-\frac{K - 1}{K} \log b \, q_{1.5}\left(\frac{k}{K - 1}\right)\right], \quad b = 500,000.$$

This retains the Geo endpoints. Both arms are full-parameter continuations from the shared checkpoint. The autoregressive probe generates exactly the tokenized passkey plus its closing marker and compares every generated token ID with that expected sequence. It does not test a subsequent terminal EOS; the EOS-complete endpoint is the separate OLMo study in Table 24.

Table 13: 750M continued-pretraining check (seed 42; 2K→ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials per length. The pooled AR row combines 2K, 4K and 8K (120 trials: Geo 80/120, EVQ-COSH 111/120).

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Answer-token exact @8K	0%	77.5%
Pooled answer-token exact (2K/4K/8K)	66.67%	92.5%

D.5 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T} = 2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ ($0.53\times$), which selects the supporting video configuration.

D.6 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ($4\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level $t=0.5$ and the same 256 evaluation videos for both arms.

Table 14 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames under bidirectional 3D RoPE.

Table 14: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bidirectional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

D.7 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed mid-point grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 4 separately measures incompatibility after a frozen post-hoc swap.

D.8 ALLOCATION UNDER GQA AND MLA COMPRESSION

A seed-42 125M study compares five attention configurations, each trained for 100M FineWeb-Edu tokens at length 2048 with 2% passkey mix. Within each configuration Geo and Cosh ($\tau = 1.414$) use the matched training recipe. Table 15 preserves the complete configuration comparison from the recorded experiment report.

Table 15: **Allocation effects across attention configurations.** Paired Geo/Cosh PPL at each length; the last column is the reported teacher-forced passkey retrieval aggregate, not autoregressive exact. One training seed per arm.

Configuration (K)	2K	4K	8K	16K	Retrieval (%)
MHA (32)	44.24/43.62	61.75/58.09	124.0/106.2	213.5/185.3	62.0/64.7
GQA-4 (32)	49.37/50.76	59.67/60.73	112.6/101.0	191.0/162.6	55.3/52.7
GQA-2 (32)	50.56/50.56	67.98/57.45	130.5/99.3	220.2/172.9	62.0/51.3
MLA-32 (16)	65.47/65.59	98.53/104.0	240.1/225.1	400.2/363.7	57.3/50.0
MLA-16 (8)	67.13/65.92	68.07/69.84	245.5/128.2	653.4/340.4	48.7/51.3

All five configurations improve long PPL, with the largest relative effect in the eight-pair MLA setting. The ordering across architectures is nonmonotone: KV sharing, latent compression, and rotary-pair count jointly define the setting. This supports studying allocation within architectural constraints, complementing the three-seed 432M experiment rather than establishing a one-variable scaling law for pair count.

E SUPPORTING RESULTS

E.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024a), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified $\tau = 1.414$ setting. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree.

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

Table 16: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond, showing that the allocation also composes with this wavelength-dependent extension.

E.2 ADDITIONAL MODEL COMPARISONS

The following comparisons provide protocol-level detail for the main results.

Table 17: **Table and amplitude separated on the Qwen-2.5-3B 36-row panel (%)**. Matching the coefficient across tables makes the table contrast comparable; the last row varies amplitude alone at a fixed table.

Table	Coefficient c in $g = 1 + c \ln s$	32K	128K
MrRoPE-Pro	0.1	87.22	78.13
BM	0.1	91.67	70.83
MrRoPE-Pro	0.074	98.33	75.35
BM	0.074	100.00	70.00
BM	$g = 1$	89.58	58.82

Table 18: **Matched six-task RULER comparisons (%)**. Each column averages six tasks ($s = 4$; short/long prompt counts per arm).

Model	Lengths	Rows	Short		Long	
			MrPro	BM	MrPro	BM
OLMo-1B	4/16K	24/48	37.85	81.81	2.78	51.32
Qwen-3B	32/128K	12/24	87.22	91.67	78.13	70.83
Qwen-7B	32/128K	6/12	83.33	80.00	84.44	71.11

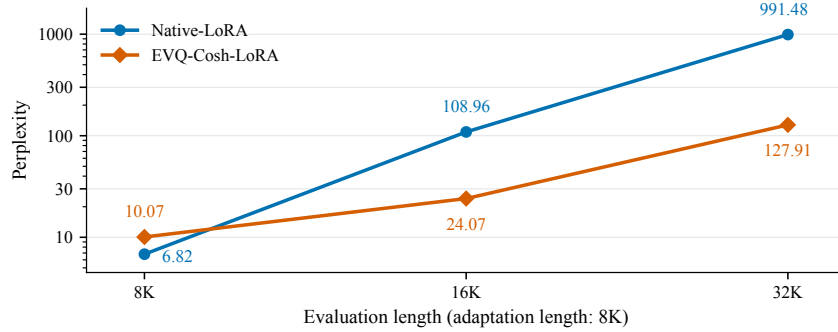


Figure 7: **Cosh allocation improves extrapolation under matched adaptation**. PPL is the exponential of equal-domain mean NLL.

Table 19: **Frozen-model RULER scores (%) at matched exponent range**. OLMo: held-out nine-task confirmation at 16K. Qwen: four-task development panel at 64K, 20 rows/task.

Exponent allocation	OLMo-1B	Qwen-1.5B
Geometric	0.56	57.75
Coarse ramp	61.04	64.00
Residual-guided	60.47	66.50

Table 20: **Static extension across native and extended windows**. Qwen-0.5B RULER-13 macro (%); g is fixed amplitude.

Installed table	g	32K	64K
Native	1.0000	54.78	22.05
YaRN-2	1.0693	55.94	45.37
Normalized-index-2	1.0513	55.92	51.46

Table 21: **Epistemic status of theoretical and empirical components.** Differentiating exact basis geometry, surrogate design criteria, heuristic operating rules, and downstream behavior.

Tier	Component	Object	Epistemic Status
1	Positional Basis	Subspace overlap $c_{\omega\nu}$, rank r_2 , limit V_0	Exact operator geometry (Prop. 1)
2	Surrogate Functional	Variational objective $\mathcal{C}_{\text{app}}[\rho]$ in (7)	Explicit design prior; not downstream $\mathcal{L}_{\text{CE}}$
3	Operating Rule	Cosh density ρ_τ and finite quantile grid	Unique density optimum; specified finite grid
4	Downstream LM	Tail NLL, RULER-13, natural LongBench QA	Measured under declared model and protocol

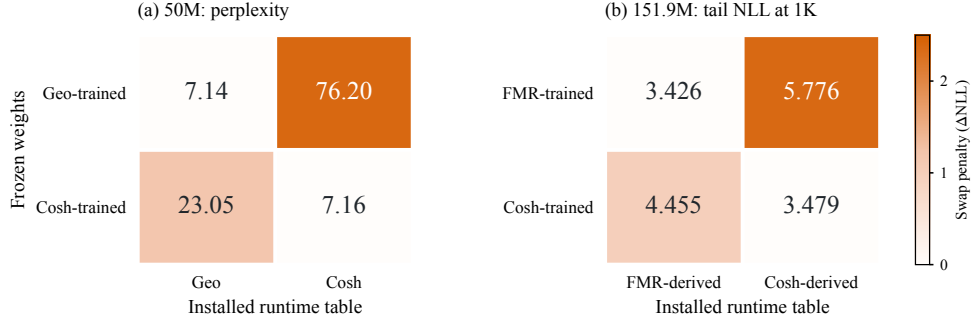


Figure 8: **Weights co-adapt to their exponent allocation.** Each row freezes one trained weight set and changes only its runtime table. Left: 50M crossing, PPL on eight windows. Right: 151.9M crossing, mean tail NLL on 32 anchors in each of two training seeds, using extensions derived from the two training tables. FMR is the geometric table at the fixed training range; shading shows the NLL increase relative to the matched diagonal cell.

F MATURE-SCALE PROTOCOLS

Table 22: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on. The Llama rows use the separate RULER-family continuation: 516 steps, rank-64 Q/K/V/O LoRA, length 8192, seed 20420726. The 300-step LongAlpaca adaptation supplies the temporal-PPL and gold-block results, not these RULER scores.

Model / protocol	Arm / metric	1 ×	2 ×	4 ×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

F.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the standard $u_k=k/K$ grid with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (Team OLMo et al., 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, with explicit position IDs providing target-range relative phases. On 8K niah_single_1, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set.

Complete numeric answers after matched routing continuation. A separate follow-up to the routing parents applies the same 100-step query-gap continuation and 32-step answer-plus-EOS continuation to Native and EVQ-COSH (training seed 20260728). Every backward pass uses at most 4096 physical tokens; explicit position IDs expose target-range phases, with maximum observed position ID 16,257. Evaluation uses real contiguous prompts from the same numeric NIAH family and greedy decoding. The endpoint requires the complete expected answer string followed by observed terminal EOS, measured from generated token IDs.

Stage identities and supervision. The starting artifacts are the two 300-step counterfactual-routing parents (rank-64, alpha-128 Q/K/V/O LoRA). Their SHA-256 prefixes are 6570ab94aec6 (Native) and 95ceeb70117c (Cosh); the later stages continue each arm’s own parent. Table 23 specifies the two matched follow-ups. Both use fused AdamW with $(\beta_1, \beta_2) = (0.9, 0.95)$, zero weight decay, gradient clipping 1, BF16, and cosine decay to 0.1 of peak LR after linear warmup. Global batch 8 uses microbatch 4 and accumulation 2. The trainable Q/K/V/O adapters contain 16,777,216 parameters; the base checkpoint is frozen.

Table 23: **The matched continuation producing complete-answer outcomes.** Training seed 20260728. Each cycle contains two counterfactual-routing updates and one natural-text update. Counts include both variants of a routing pair.

Stage	Steps	Peak LR / warmup	Supervision and position policy
Query-gap	100	$5 \times 10^{-5} / 20$	Answer CE plus counterfactual margin; semantic query-block offsets through 16K.
Answer + EOS	32	$10^{-5} / 4$	Answer CE, immediate EOS CE (weight 1), same counterfactual margin; realized source-to-answer gap curriculum.

Routing examples contain eight key-value blocks and one query. The two variants replace the queried source value while preserving the surrounding context; the answer changes accordingly. The training set has 1024 paired examples, the separate calibration set 128, with seeds 20293725 and 20304725 and different query templates. The final stage reuses the raw training rows with rebuilt answer-plus-EOS labels, covering three answer tokens and EOS; padding and other prompt tokens carry no answer loss. The counterfactual term uses margin 1 and weight 0.5, comparing the correct and swapped answer where their tokens differ. Natural replay uses ordinary next-token loss. The 100-step stage contains 67 routing/33 natural updates; the final stage contains 22/10.

Position-ID construction. Let q be the semantic query-block start, a the answer start, h the end-exclusive source span, and n the active sequence length. Context positions before q remain contiguous. Both counterfactual variants share the position array:

$$p_i = i + o \mathbf{1}\{i \geq q\}.$$

In the 100-step stage, each routing update uses four pair offsets: a low offset alternating between 0 and $[1, 4096]$, one from $[4097, 8192]$, and two from $[8193, 12289]$. Values traverse each band by a seed-derived coprime modular stride, with a deterministic per-update permutation. In the final stage,

the stream instead specifies realized gap targets: the low target alternates between contiguous and [4096, 8191], one target lies in [8192, 12287], and two in [12288, 16079]. For a noncontiguous target g , set

$$o = \min\{g - (a - h), 12289, 16384 - \min(n, 4095)\}.$$

The implementation requires the resulting gap to remain in the target band and all active position IDs below 16384. Natural replay remains contiguous. The exact two deterministic schedules are included in `figs/verify_routing_schedule.py`; recomputation matches the run-record hashes for all 268 query offsets and all 88 final-stage gap targets. This procedure exposes target-range phases during training while keeping physical sequences at most 4096 tokens.

The evaluation uses separately generated, real contiguous numeric-NIAH prompts, 100 per length, with greedy decoding and complete-string-plus-EOS scoring. Training uses custom counterfactual query templates rather than the official evaluation template. The source receipts and full parent/final adapter hashes are included in `figs/routing_protocol_receipts.json` and `figs/recovered_asset_inputs.json`.

Table 24: Complete-answer-string plus terminal-EOS exact match after the matched routing follow-up, 100 inputs per length and one trained pair.

Length	Native	EVQ-COSH
4K	95/100	100/100
8K	18/100	98/100
16K	0/100	60/100

All final generations in both arms terminate with EOS. On the EVQ-COSH subsets whose source-query gap exceeds the original training range, exact match is 48/50 at 8K and 31/66 at 16K. These outputs belong to the additional continuation above, separately from the earlier first-number retrieval comparison.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled Q/K `metrics.json` records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 22 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals quantify evaluation-row variation for the trained pair.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/ +0.0381/ -0.0437/ -0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K.

The EVQ-COSH branch uses a Hugging Face single-GPU loop; the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch,

4K context, 1,000 optimizer steps, and 2,097,152,000 counted input tokens. This comparison includes the frequency-table change and the two trainer implementations.

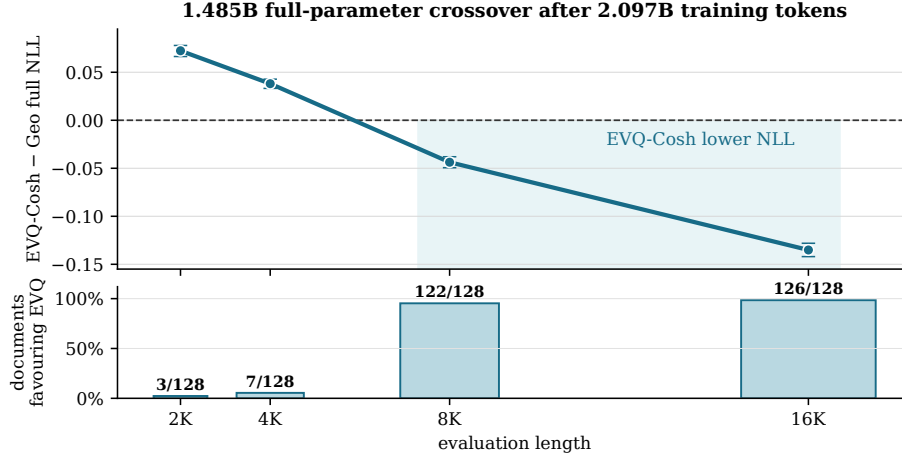


Figure 9: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

F.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Table 19 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The reference rows use the original Native table and the Transformers YaRN frequency map. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm, $\tilde{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$; it sets $w_k = (1 - \tilde{U}_k)^2$ in the frequency blend of Eq. (54), so less unique pairs move farther.

Numerical construction of the fixed-support controls. The following specifies the derived arm independently of the later FullLagP2 construction. Use the checkpoint’s actual float32 native tensor promoted to float64, with $L = 4096$ for OLMo and 32768 for Qwen. Start from integer separations $d = 0, \dots, L - 1$ and weights $L - d$. Group consecutive separations in blocks of two, retain each block’s rounded weighted-centroid separation, and sum its weights; normalize the weights to unit mass. This gives 2048 and 16384 weighted support points, respectively.

Build X with columns $\sqrt{w_d} \cos(d\omega_k)$ and $\sqrt{w_d} \sin(d\omega_k)$ in float64. For each two-column block X_k , solve $B_k = \arg \min_B \|X_k - X_{-k}B\|_F$ with singular-value cutoff `rcond` = 10^{-10} , and compute

$$U_k = \text{clip}\left(\frac{\|X_k - X_{-k}B_k\|_F^2}{\|X_k\|_F^2}, 0, 1\right).$$

If the marginal energy is zero, set $U_k = 0$. Min-max normalize the U_k values; if their span is zero, use the zero vector. Then $w_k = (1 - \tilde{U}_k)^2$ and $\omega'_k = \omega_k(1 - w_k) + \omega_k w_k/4$. Cast to float32, overwrite the first frequency with the exact native first entry and the last with the float32 native last entry divided by four, and require strict positivity and decreasing order. The resulting tensor SHA-256 prefixes are a435d7544144 (OLMo) and 3512335e408c (Qwen).

The uniform control is log-linear between the same endpoints. The coarse control enumerates all integer ramp bounds $0 \leq l < h < K$, minimizes mean squared distance to w_k in movement space, and breaks ties lexicographically in (l, h) . It uses the same frequency blend and endpoint overwrite. The selected bounds are (20, 22) for OLMo and (28, 31) for Qwen. FullLagP2 in Appendix G.9 instead uses every integer separation and installs a log-frequency shift; its measurements are a separate experiment.

Table 25: **Exact frozen mature-checkpoint RULER macro (%)**. These are the values plotted in Table 19. Reference rows are separated from the fixed-support block; only the lower block changes interior z while holding the stated checkpoint protocol fixed.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalue, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are $[54.88, 64.80]$ points for OLMo and $[0.25, 17.50]$ points for Qwen; the paired derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

Fresh-distribution allocation and routing controls. On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 26 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior z changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 26: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

Contrast	8K	16K
Session-s4 – Native	−4.0953	−4.4393
Geometric – derived z	−0.0288	+4.4560
Coarse ramp – derived z	+0.0005	+0.0006

The OLMo deployment policy selects the exact Native table when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at $2\times$ and 24.97/25.59% at $4\times$. The QA arms use identical rows, and every post-chat input plus generation reserve fits within 16K. Table 19 gives the corresponding allocation-only comparison.

Table 27: Compact zero-training deployment endpoints on the frozen OLMo checkpoint, using the specified table, amplitude, and session routing.

Endpoint	Official YaRN ($s=4$)	Zero-training policy
PG-19 tail NLL, 8K/16K	3.4404/3.7955	3.1060/3.0977
RULER-13 macro, 8K/16K (%)	23.82/5.88	67.72/54.40
Qasper token F1, 16K ($n=200$, %)	18.03	24.57
2Wiki token F1, 16K ($n=200$, %)	25.69	26.66

F.3 LLAMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table.

Probability and causal source use. The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate 10^{-4} . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. Native uses the pretrained endpoint grid, and Cosh uses midpoint quantiles; the same adaptation recipe is applied to these two frequency tables. The frozen adaptation tensor is drawn from LongAlpaca-12k (Chen et al., 2023b): 7,476 training and 152 validation rows after length filtering, with effective batch 8 (microbatch 2, accumulation 4).

Temporal holdout and aggregation. Evaluation uses arXiv titles/abstracts, Federal Register titles/abstracts, and Stack Overflow question bodies selected by publication or creation timestamps between January 1 and July 12, 2026. These are separate from the LongAlpaca adaptation corpus. Collection was frozen before evaluation; each domain has eight disjoint-document 32,768-token packs formed by concatenation with EOS separators. Positions and causal attention continue across document boundaries. The 8K and 16K conditions are exact prefixes of the same 32K packs. All ordinary text targets in each prefix are scored, excluding document-first tokens and inserted EOS separators. For domain d , let $\ell_d(L)$ and $n_d(L)$ be the summed token loss and valid-token count over its eight packs. The reported aggregate is

$$\text{NLL}(L) = \frac{1}{3} \sum_{d=1}^3 \frac{\ell_d(L)}{n_d(L)}, \quad \text{PPL}(L) = \exp(\text{NLL}(L)).$$

Table 28 gives token counts and the distribution of the 24 paired pack-level NLL differences. The accompanying `figs/llama_temporal_summary.json` retains each pack’s loss sum and count, its domain identity, and the frozen collection hash.

Table 28: Llama-3-8B temporal holdout: scored tokens and the 24 paired pack NLL differences (EVQ-COSH minus Native). Each length uses the same packs.

Prefix	Valid tokens/arm	Valid tokens/pack	Pack ΔNLL : min / median / max
8K	194,794	8,051–8,183	+0.193/ + 0.432/ + 0.613
16K	389,672	16,111–16,353	−2.069/ − 1.436/ − 1.278
32K	779,400	32,243–32,679	−2.319/ − 2.074/ − 1.754

The domain-macro EVQ-COSH-minus-Native NLL is +0.390/ − 1.510/ − 2.048 at 8/16/32K.

Position-preserving source intervention. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry. NLL pools the negative log probabilities of the 30 answer tokens (3 per case); deletion changes this mean by −0.0095 for Native and +1.5055 for EVQ-COSH. The EVQ-COSH change is positive in all ten cases. Case-level hit@16 ranges are

3.12–34.38% for Native and 62.50–68.75% for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

Generation and natural-QA outcomes of the same adapters. The source interventions reveal how remote evidence affects answer probabilities. Task evaluation measures whether those probabilities produce the required output. In the seed-42 capability evaluation, Cosh passkey exact is 100/0/0% at 8/16/32K, and single-needle top-1 is 56.67/3.33/0%. The registered three-arm QA comparison gives task-macro F1 0.2309 for the base model, 0.2110 for Native-LoRA, and 0.1126 for Cosh-LoRA; the paired Cosh-minus-Native difference is -0.0984 , with 95% bootstrap interval $[-0.1297, -0.0697]$. The deficit is concentrated in inputs of at most 8K; on the longer subset all arms are near the floor. Forced-gold inclusion changes NLL by -0.034 , and the sparse readout pilot does not improve conversion. These report-backed outcomes accompany the probability and source-use measurements of the same 300-step adapters. They motivate distinguishing access to a distant source from successful task readout.

RULER-family continuation. A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 22 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

G FROZEN EXPONENT-ADJUSTMENT PROTOCOLS

This section specifies the static profiles and evaluation panels used in §6. An amplitude g multiplies both the cosine and sine tables, so its contribution to a rotated Q/K logit is g^2 . Every table is installed before prefill and retained for the entire request. The accompanying `figs/recorded_runtime_identities.json` records checkpoint, tokenizer, and evaluator identities. The index/placement studies use RULER commit `c3f5e3b4f87f97e048793bb510a3a6b19a46bf3a`.

For the native-relative displacement in (11), the three installation laws are

$$d_k = \begin{cases} -\log(1 - w_k + w_k/s), & \text{frequency blend,} \\ m_k \log s, & \text{log-frequency shift,} \\ \sum_{j < k} \log \lambda_j, & \text{mixed-radix conversion.} \end{cases} \quad (54)$$

The blend weight w_k and log-shift profile m_k have different meanings; the first line is the exact conversion between them.

G.1 NORMALIZED-INDEX PROFILE

For a native table with K pairs, base b , and reference length L_{ref} , the normalized-index construction evaluates the profile on a 64-pair reference grid with the same b, L_{ref} :

$$\bar{\omega}_j = b^{-j/64}, \quad c_{64} = (1 - b^{-1/64})^{-1}, \quad \xi_j = \log \frac{L_{\text{ref}} \bar{\omega}_j}{2\pi c_{64}}, \quad (55)$$

$$\bar{m}_j = \text{clip} \left(\frac{\xi_H - \xi_j}{\xi_H - \xi_L}, 0, 1 \right), \quad \xi_H = 0.7382780681078285, \quad \xi_L = 0.366403835112904. \quad (56)$$

These two constants were fitted to the existing OLMo displacement profile and then frozen. Linear interpolation of $(j/63, \bar{m}_j)$ at $k/(K-1)$ gives m_k on the target grid. The installed table is $\omega'_k = \omega_k^N s^{-m_k}$.

The Qwen-2.5-0.5B-Instruct experiment uses its native $K = 32$ pairs, $b = 10^6$, and original context length $L_{\text{ref}} = 32768$, with $s = 2$ and fixed amplitude $g = 1 + 0.074 \log 2 = 1.0512928914$.

The YaRN-2 comparison uses the Transformers official-equation table and $g = 1 + 0.1 \log 2 = 1.0693147181$; Native uses $g = 1$.

The full RULER-13 confirmation uses seed 202609027, 20 examples per task and length, the same tokenizer and native chat template, and greedy task-budgeted generation. It contains 520 generations per arm across 32768 and 65536 tokens. Per-task scores are shown in Table 29. The macro averages the thirteen task means equally. Paired intervals use 10,000 bootstrap samples within task strata with seed 202609028. The 64K index-minus-YaRN difference is 6.0897 percentage points, with interval [2.7627, 9.5835]; at 32K it is -0.0256 points, with interval $[-3.2630, 3.2179]$.

Table 29: Qwen-0.5B RULER-13 per-task score (%), 20 examples per cell.

Task	Native	32K		Native	64K	
		Index	YaRN		Index	YaRN
Single-1	100.00	100.00	100.00	75.00	100.00	100.00
Single-2	100.00	100.00	100.00	15.00	100.00	75.00
Single-3	100.00	100.00	100.00	30.00	100.00	95.00
MultiKey-1	90.00	95.00	90.00	50.00	90.00	65.00
MultiKey-2	50.00	60.00	35.00	0.00	40.00	25.00
MultiKey-3	5.00	0.00	10.00	0.00	5.00	5.00
MultiValue	62.50	81.25	81.25	22.50	63.75	58.75
MultiQuery	62.50	80.00	77.50	22.50	65.00	50.00
Variable tracking	48.00	42.00	46.00	3.00	28.00	36.00
Common words	2.50	2.00	2.50	7.00	0.50	0.00
Frequent words	46.67	31.67	35.00	31.67	46.67	40.00
QA-1	20.00	10.00	30.00	5.00	15.00	20.00
QA-2	25.00	25.00	20.00	25.00	15.00	20.00

G.2 PLACEMENT ON ANOTHER FINITE PAIR GRID

The alternative direct-gap construction evaluates the same clipped profile using the target table’s local frequency spacing:

$$c_K = (1 - b^{-1/K})^{-1}, \quad \xi_k^{(K)} = \log \frac{L_{\text{ref}} \omega_k^N}{2\pi c_K}, \quad m_k^{\text{gap}} = \text{clip} \left(\frac{\xi_H - \xi_k^{(K)}}{\xi_H - \xi_L}, 0, 1 \right).$$

Both placements share the frozen ξ_H, ξ_L and amplitude coefficient 0.074. Within each comparison, endpoints, scale, amplitude, checkpoint, inputs, decoder, and scorer are matched. Table 30 reports two independent-input confirmations with 80 rows per task.

Qwen-0.5B uses the 32-pair, $s = 2$ setting above and data seed 202609026, with four tasks at each of 32K and 64K. These are separate inputs from the subsequent RULER-13 confirmation. Gemma-1.1-2B-Instruct uses $K = 128$, $b = 10,000$, a fixed operating reference of 4096, $s = 4$, and $g = 1 + 0.074 \log 4 = 1.1025857827$. Its confirmation uses data seed 202609028 at 16K. The task vector is NIAH single-1, multikey-2, multikey-3, and variable tracking. Gemma’s direct-gap scores are 100/88.75/20/82.50%; normalized-index scores are 100/88.75/40/87.25%. Both tables were frozen before these inputs were generated, and the earlier selection rows are excluded. This confirmation compares the two placements; the Native/YaRN baseline panel uses a different, earlier input set.

Table 30: Independent-input placement comparisons, 80 examples per task and four tasks per panel. Intervals are for index minus direct-gap, in percentage points: 97.5% for each Qwen length (Bonferroni over two lengths), 95% for Gemma.

Model	Length	Direct-gap (%)	Index (%)	Difference interval
Qwen-0.5B	32K	49.69	53.37	$[-0.25, +7.75]$
Qwen-0.5B	64K	46.62	46.13	$[-4.88, +3.81]$
Gemma-2B	16K	72.81	79.00	$[+2.81, +9.63]$

G.3 BOUNDARY-MATCHED INTERMEDIATE EXPONENTS

Let l be the last native frequency pair making more than 32 turns over the reference window and h the first making less than one turn. Write $N = h - l$ and $q = \text{clip}(k - l, 0, N)$. All profiles leave the fast band unchanged and divide the slow-band frequencies by s :

$$\omega'_k = \omega_k^N s^{-m_q}, \quad m_0 = 0, \quad m_N = 1. \quad (57)$$

With the same band boundaries, the three intermediate allocations are

$$m_q^{\text{Uni}} = \frac{q}{N}, \quad (58)$$

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad (59)$$

$$m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (60)$$

The Uni and Pro rules follow the uniform and progressive mixed-radix constructions of Tian et al. (2026). BM smooths the radix increments $\epsilon_q = m_q - m_{q-1}$ at both band boundaries.

To obtain BM, minimize $\sum_{q=0}^N (\epsilon_{q+1} - \epsilon_q)^2$ with $\epsilon_0 = \epsilon_{N+1} = 0$ and $\sum_{q=1}^N \epsilon_q = 1$. The discrete Euler equation gives

$$\epsilon_q = \frac{6q(N+1-q)}{N(N+1)(N+2)};$$

summing gives (60). The quadratic form is strictly convex with the endpoints fixed, so this solution is unique. Figure 10 shows the actual profile shapes for the tested transition widths.

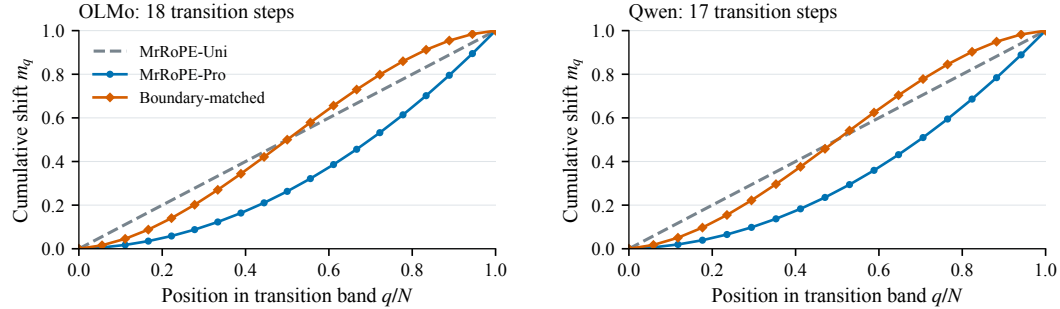


Figure 10: **Intermediate-band exponent displacements.** All three profiles have the same cumulative scale at the endpoints. BM redistributes the increments toward the middle of the transition band.

OLMo-2-0425-1B-Instruct uses $b = 500,000$, reference length 4096, $K = 64$, and $(l, h, N) = (14, 32, 18)$. Qwen-2.5-3B/7B-Instruct use $b = 10^6$, reference length 32768, $K = 64$, and $(l, h, N) = (23, 40, 17)$. The factor-four comparisons use $g = 1 + 0.1 \log 4 = 1.1386294361$ for both MrRoPE-Pro and BM, with FP32 frequency tables and BF16 model execution.

G.4 SIX-TASK MODEL COMPARISONS

The shared task types are NIAH single-2, multikey-2, multiquery, variable tracking, frequent words, and QA-1. OLMo’s confirmation uses seed 20260910 with four short and eight long examples per task, after a separate development panel with seed 20260909. Its 4K/16K scores are listed in Table 18; MrRoPE-Uni on the same confirmation inputs scores 76.88/32.12%, and official YaRN scores 54.38/6.94%. The Qwen-3B panel has two 32K and four 128K examples per task. The 7B screen reuses the first one short and two long examples per task; this selection was fixed before its outputs were inspected. Table 31 reports every task mean.

All arms use the corresponding native chat template and the same task-specific generation caps, score the entire saved response with the RULER task metric, and retain generated token IDs and EOS flags. On Qwen-7B, a tokenwise MLP chunk size of 4096 makes 128K evaluation fit in memory; both arms share that path while retaining full attention.

Table 31: All six-task BM/MrRoPE-Pro comparisons. Values are official RULER task scores (%); model-specific sample counts are given in Table 18.

Model	Task	Short		Long	
		MrPro	BM	MrPro	BM
OLMo-1B	Single-2	75.00	100.00	0.00	100.00
	MultiKey-2	25.00	100.00	0.00	37.50
	MultiQuery	18.75	87.50	0.00	43.75
	Variable tracking	25.00	70.00	0.00	10.00
	Frequent words	58.33	58.33	16.67	41.67
	QA-1	25.00	75.00	0.00	75.00
Qwen-3B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	75.00	50.00
	MultiQuery	100.00	100.00	93.75	100.00
	Variable tracking	90.00	100.00	75.00	75.00
	Frequent words	83.33	100.00	75.00	75.00
	QA-1	50.00	50.00	50.00	25.00
Qwen-7B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	100.00	100.00
	MultiQuery	100.00	100.00	100.00	100.00
	Variable tracking	100.00	80.00	90.00	60.00
	Frequent words	100.00	100.00	66.67	66.67
	QA-1	0.00	0.00	50.00	0.00

G.5 NATURAL QUESTION ANSWERING

The natural comparison reuses frozen, source-verified LongBench prompts (Bai et al., 2024). Inputs are untruncated, retain their source questions and references, and fit the 16K input-plus-generation budget. Selection follows the pre-existing source-hash order; no generated answer participates in selection. The complete eligible pool contains 778 questions, of which 631 have more than 4096 input tokens. Both methods retain the same factor-four table at every length.

Greedy generation uses a 32-token cap for HotpotQA and 2WikiMQA, 128 for Qasper and NarrativeQA, and 64 for MultiFieldQA-en. The metric is token F1 on the complete saved generated response against the task references. Table 32 gives the long-input and original-window-length strata, including EOS counts. The long-input macro averages the five task means; the shorter-input macro averages its four available tasks. The respective BM-minus-MrPro changes are +3.8194 and +2.0028 points, with paired within-task bootstrap intervals [1.3161, 6.2920] and [-3.5618, 7.4742]. The MultiFieldQA long-input row includes 40 capped responses in each arm; these remain in the F1 calculation.

Table 32: Natural QA under static $s = 4$. Token F1 uses the whole generated response; EOS counts are recorded separately.

Input length	Task	n	MrPro F1 (%)	BM F1 (%)	MrPro EOS	BM EOS
4K–16K	HotpotQA	166	28.63	33.54	160	163
	2WikiMQA	173	21.55	25.43	169	171
	Qasper	119	13.94	19.06	119	119
	NarrativeQA	61	9.51	11.48	61	61
	MultiFieldQA	112	34.49	37.72	72	72
$\leq 4K$	HotpotQA	5	58.33	51.67	5	5
	2WikiMQA	26	34.62	44.98	26	26
	Qasper	78	26.16	28.14	78	78
	MultiFieldQA	38	50.94	53.27	35	33

G.6 COMPACT DESCRIPTIONS OF A USEFUL ALLOCATION

An ordered 64-slot movement profile can be approximated by a compact law without retaining every local feature. Define $c_k = L\omega_k/(2\pi)$, $c_{\text{orth}} = (1 - b^{-1/K})^{-1}$, and $\xi_k = \ln(c_k/c_{\text{orth}})$. The adjacent-frequency separation is exactly $L(\omega_k - \omega_{k+1})/(2\pi) = e^{\xi_k}$. A clipped-affine law fitted to the frozen OLMo movement vector is

$$G_4(\xi) = \text{clip}\left(\frac{\xi_H - \xi}{\xi_H - \xi_L}, 0, 1\right), \quad \omega'_k = \omega_k 4^{-G_4(\xi_k)}.$$

The fitted boundaries are $\xi_H = 0.7382780681$ and $\xi_L = 0.3664038351$. The two parameters were frozen before examining Qwen geometry or model results. Both checkpoints use the fixed cosine/sine gain $1 + 0.074 \ln 4$. The fit has movement MAE 0.001223 on OLMo. The original OLMo-to-Qwen transport interpolates the full profile in $u = e^\xi$; C2 evaluates the frozen fitted law directly in ξ .

Table 33: **A compact profile preserves long-context behavior.** Official task macro (%). OLMo uses RULER-13; Qwen uses four core tasks. Long results are post-gate diagnostics with the already frozen tables; Qwen 32K was collected after the long results.

Model	Endpoint	Full 64-point profile	C2
OLMo	8K	66.71	66.84
OLMo	16K	49.86	50.24
Qwen	32K	70.00	71.25
Qwen	64K	67.25	67.75
Qwen	128K	54.50	57.25

The long-end results show that detailed 64-point structure is not required to preserve these measured behaviors. At the native window, OLMo’s PPL retention is 0.870971 and task retention 0.904932; Qwen native-task macro is 82.00% versus 71.25% for C2. The registered deployment requirement was retention at least 0.875, so the single-table deployment gate failed. This operating verdict and the positive low-dimensional long-behavior result answer different questions. The original profiles, C2, and the later normalized-index placements retain distinct tensor identities.

G.7 AN INTERIOR-PROFILE CONTROL AT FIXED TOTAL DISPLACEMENT

For $K = 64$, let $k = 1, \dots, 18$, $p_k = 6k(19 - k)/(18 \cdot 19 \cdot 20)$, and $v_k = k - 19/2$. Two increment profiles are

$$\epsilon_k^A = p_k \left(1 - \frac{10}{119} v_k\right), \quad \epsilon_k^B = p_k \left(1 - \frac{10}{119} v_k + \frac{35}{1496} \left(v_k^2 - \frac{357}{20}\right)\right).$$

Both use $m_j = 0$ through slot 14, cumulative increments on slots 15–32, and $m_j = 1$ thereafter. They have the same increment mass and centroid, identical sampled endpoints, and $\sum_j m_j = 42$. Installed frequencies are $\omega'_j = \omega_j 4^{-m_j}$ with fixed gain $1 + 0.1 \ln 4$ and terminal shift $m = 1$. This is a controlled comparison within that displacement setting.

On 350 shared 16K OLMo development inputs, scores are 43.6714% and 54.4000%. The paired difference is +10.7286 points, with 75 wins, 26 losses, and 249 ties; the paired standardized mean is $t = 5.47$. A separate 16-document continuous-text evaluation reports NLL 2.9417 versus 2.8309. These observations distinguish two internal profiles that share their total displacement. They are not a held-out method-selection win: the task panel participated in profile exploration. The numerical construction, row-score aggregation, and common-total checks are reproduced by `figs/verify_recovered_assets.py`.

G.8 FOUR METHODS ON THE SAME OLMo TASK PANEL

The six-task panel contains 24 prompts at 4K and 48 at 16K. All four methods use $s = 4$ and gain $1 + 0.1 \ln 4$. MrRoPE profiles are local formula implementations; official YaRN denotes the channel-index ramp implementation. Table 34 is recomputed from the recorded per-task aggregates, with matched BM/control comparisons recorded on the shared row identities.

Table 34: **Intermediate-profile comparison on the OLMo six-task panel.** Official task macro in percent, including partial-credit tasks.

Method	4K	16K
MrRoPE-Pro(4)	37.85	2.78
Official YaRN(4)	54.38	6.94
MrRoPE-Uni(4)	76.88	32.12
BM(4)	81.81	51.32

The official YaRN and Uni raw-output files have recorded hashes but are not bundled with the source archive; their per-task aggregates and identities are preserved. The separate natural-QA experiment in Appendix G.5 evaluates the BM/MrRoPE-Pro comparison on natural inputs.

G.9 FULL-LAG POSITIONAL GEOMETRY AS A CANDIDATE CONSTRUCTION

FullLagP2 constructs the native sine-cosine column pairs across all separations $d = 0, \dots, 32767$ with causal weights proportional to $32768 - d$. For each pair it computes its least-squares residual fraction relative to the other pairs (cutoff 10^{-10}), min-max normalizes these fractions to $\bar{u}_j$, and sets $m_j = (1 - \bar{u}_j)^2$. The resulting full table uses $s = 4$ and gain $1 + 0.074 \ln 4$. It depends on positional geometry, with no model-weight or activation fitting.

Table 35: **Recorded FullLagP2 task outcomes on Qwen-1.5B.** Eight prompts/task/length, official task score in percent. The matched-gain MrPro arm uses the same gain as FullLagP2.

Length	Task	MrPro	MrPro, matched gain	FullLagP2
64K	MultiKey-2	12.50	25.00	37.50
	Variable tracking	82.50	77.50	87.50
	Frequent words	45.83	45.83	70.83
128K	MultiKey-2	0.00	–	0.00
	Variable tracking	72.50	–	85.00
	Frequent words	50.00	–	50.00

This is a positive small-panel example of converting full-pair geometry into a useful allocation. With the same full array on Qwen-3B at 64K (four prompts/task), FullLagP2/MrPro scores are 75/50% for MultiKey-2, 90/95% for variable tracking, and 66.67/75% for frequent words. The decoder uses repetition penalty 1.1; these outputs remain separate from later screening panels. The paired score fractions and checkpoint identities are preserved in `figs/recovered_asset_inputs.json`. The construction’s geometric residual is a design statistic, and the task measurements establish where it is useful.

G.10 PHASE GEOMETRY AND MEASURED PROFILE RESPONSES

For $s > 1$, the installed frequency $\omega'_j = \omega_j s^{-m_j}$ maps separation Δ to native effective separation Δs^{-m_j} . At $\Delta = sL_{\text{train}}$, containment within the unwrapped training interval requires $m_j \geq 1$. This is a per-pair geometric condition: distinct m_j generally map to distinct native separations, rather than one shared training position. The phase sensitivity at the native table is

$$\left. \frac{\partial(\Delta \omega_j s^{-m_j})}{\partial m_j} \right|_{m_j=0} = -\Delta \omega_j \ln s.$$

It quantifies local phase change; the finite drift at $m_j = 0$ is zero.

Proposition 1 characterizes positional directions, while the learned content coefficients can differ across pairs. We assess deployment profiles through complete-generation task scores.

For twelve OLMo-1B profiles on 350 shared 16K RULER inputs, a count of slots whose test-distance turns lie in $[0.25, 16]$ separates three response groups. Appendix G.11 defines the count, tabulates every arm, and reports its reconstruction provenance. The count is an exploratory summary of this panel: it separates between-group scores while leaving within-group rankings unresolved.

G.11 RECONSTRUCTING THE TWELVE-PROFILE RESPONSE ANALYSIS

Statistic and evaluation unit. For OLMo-2-0425-1B-Instruct, $K = 64$, $b = 500,000$, native window 4096, and test distance $D = 16384$, define

$$T_j(D; m) = \frac{D}{2\pi} b^{-j/K} 4^{-m_j}, \quad N(m) = \sum_{j=0}^{63} \mathbf{1}\{0.25 \leq T_j(D; m) \leq 16\}. \quad (61)$$

This is a band-count statistic. The thresholds are exploratory choices examined on the same twelve profiles, not calibrated-readout boundaries. All arms use $g = 1 + 0.1 \ln 4 = 1.1386294$, 350 shared row identities, seven equally sized RULER task groups, and 16K inputs. The score is the mean saved task score in $[0, 1]$, including partial credit; EOS is recorded separately. The 350 inputs estimate each profile’s task score, whereas the configuration-level analysis has twelve profiles. MrPro and BM use archived endpoint outputs; the other ten arms have separate saved outputs.

Table 36: **All twelve profiles used for exploratory response fitting.** Scores are recomputed from saved row-level outputs. SHA prefixes identify reconstructed little-endian float32 frequency tensors; these are reconstruction digests, not recovered runtime tensor attestations. All twelve profiles participate in the fit.

Profile	N	Score sum	Mean (%)	EOS / 350	SHA-256 prefix
MrPro	15	24.80	7.09	197	505d1fb2dce3
beta_b0p25	16	50.65	14.47	181	a8313cac3ee8
turns_a0p5b32	16	56.15	16.04	169	25721495d712
beta_b0p5	16	81.20	23.20	150	4b0b97569a3a
turns_a1b16	16	92.80	26.51	167	a24e3329fc49
knife_both	17	134.60	38.46	173	4c9d6fa7684b
knife_taper	17	137.80	39.37	146	a3b8138d159e
knife_leak	17	138.90	39.69	120	8f92ffecc376
MrProBM	17	145.85	41.67	119	0ed902cba69b
beta_b2	17	175.05	50.01	119	1188a7d3c34c
turns_a2b32	17	178.50	51.00	129	cbd8c451c0bb
turns_a1b64	17	188.45	53.84	130	7aedd80ef093

Fit and scope. Unweighted least squares of the logit-transformed mean score on N gives $\text{logit}(\bar{r}) = -20.45 + 1.19N$. Among 66 profile pairs, 39 have different counts and concordant scores, 0 have different counts and discordant scores, and 27 tie in count. The count therefore describes between-group ordering and does not resolve within-group differences. This is a descriptive fit to fractional scores, not a binomial success model or an independent predictive test. Threshold choices and coefficients were analyzed using these same profiles.

Reconstruction and provenance. The accompanying `figs/profile_diagnostic_inputs.json` records every m_j , full reconstruction digest, row-output digest, score sum, and fit coefficient. The common row-identity digest is `14736f284ac30c0c`. Matching row identities checks panel alignment in the saved outputs; the available records do not supply a new prompt-byte comparison for the archived endpoints. The analysis therefore reports the response of the recorded profiles.

Testing the interpretation beyond the fitted family. A subsequent matched continuous-text evaluation at 16K on 16 documents increased the count from $N = 17$ for BM to 22–26 by releasing slow-end interpolation. BM had NLL 2.8627; the five modified tables gave 2.8971, 2.8674, 2.8697, 2.8878, and 3.0372. Every modification increased the count while worsening this likelihood endpoint. The task-score fit above and this independent likelihood instrument measure different outcomes; together they show why the geometric count alone is insufficient to select a deployed table. Positional coverage remains a computable description, and its usefulness must be established through the learned model and the target task.

The archived BM endpoint is also not an exact text-replay target: a later same-name rerun differed on 24/350 row scores and 210/350 generated texts despite tiny reconstructed tensor differences. Table 36 reports the saved archived outputs; it does not pool them with reruns.